

Geofenced Demand-Responsive Transit: Autonomous Vehicle Fleet Sizing and District Design Under Spatial Demand Uncertainty

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Highlights

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- The main contribution of this paper lies in developing an autonomous-vehicle-based, demand-responsive service-region design framework that jointly optimizes depot location, geofenced spatial partitions, fleet sizing, and dispatch intervals, and incorporates feature-based Operational Design Domain (ODD) costs that reflect the autonomy requirements of operating autonomous vehicles (AVs) in heterogeneous road environments.
- We quantify how heterogeneous road environments affect AV deployment by converting roadway features into ODD costs, and embed these feature-based ODD costs into the joint optimization to ensure districts with more complex environments incur appropriately higher autonomy requirements.
- We address the ambiguous demand patterns by developing a Wasserstein distributionally robust formulation that ensures balanced workloads, consistent service levels, and resilience to shifting demand patterns.
- The practical value of our framework is demonstrated through extensive case studies based on real service data, showing substantial reductions in operational cost, and shorter user wait times.

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ABSTRACT

We propose a class of last-mile, multi-passenger autonomous vehicle (AV) transportation service design problems, motivated by the shortage of human drivers or mobility options in distributed regions during off-peak hours. In such settings, the autonomous nature of AVs provides significant cost and reliability advantages by eliminating driver expenses and operating in infrequently visited districts. Typically, efficient deployment of AVs requires (i) the concept of a geofence, defining the spatial boundary within which each type of AV can safely operate, and (ii) the operational design domain (ODD) cost, representing the training and adaptation expenses required for AVs to handle different driving environments. Due to these considerations, policymakers prefer fixed and stable service partitions with geofenced boundaries to simplify long-term planning, yet these partitions can easily become misaligned when spatial demand patterns shift or drift over time. To this end, we propose a generalizable *spatial-temporal demand batching* model for AV-based last-mile services that jointly optimizes district partitioning, fleet sizing, and dispatch scheduling by addressing spatial demand ambiguity through Wasserstein distributionally robust optimization. We model the ODD cost through a modular structure that captures how AV capabilities are fine-tuned for specific district features such as rivers, bridges, and road complexity. We then develop a discretization method which provides a new angle to reformulate our optimization problem, resulting in neat convex inner programs and computationally tractable solution methods even if we have complex modeling considerations. The proposed framework is also applied to a nonprofit last-mile transportation service provider in East Pittsburgh to demonstrate its superior performance to the current practice based on fixed routes. Overall, the proposed framework establishes a reliable and efficient approach for deploying AV-based last-mile services under spatial demand ambiguity and practical ODD constraints. We highlight that our convex reformulation techniques are agnostic of the nuances of our engineering setup, and could be reusable for other variants of spatial-temporal demand batching problems.

1. Introduction

Last-mile service, connecting major hubs to final destinations (Wang and Odoni, 2016; Shehadeh, Wang and Zhang, 2021), is recognized as one of the most costly and operationally complex segments of logistics and urban mobility (Silva, Amaral and Fontes, 2023). This segment contributes disproportionately to congestion and emissions because of frequent stops, short-haul trips, and dense urban operations (Bachofner, Lemardelé, Estrada and Pagès, 2022; Silva et al., 2023; Engesser, Rombaut, Vanhaverbeke and Lebeau, 2023). The increasing deployment of Autonomous Vehicles (AVs) introduces opportunities for substantial improvements in efficiency, safety, and sustainability in this critical segment (Fagnant and Kockelman, 2015; Engesser et al., 2023). However, determining how AV-enabled last-mile services should be structured (e.g. depot placement, fleet allocation, and spatial service region design), remains an open challenge due to autonomy-specific constraints.

Recent AV deployments in both passenger and freight services, including Waymo's rider-only taxis (Waymo LLC, 2025), TIER IV's Autoware-based pilots (Kutuk, 2022), and Nuro's driverless delivery operations (Korosec, 2020), demonstrate that autonomous operation is already feasible at commercial scale. For example, Autoware, the open-source AV stack developed by TIER IV, has been used by more than 100 companies and runs on more than 30 vehicles in over 20 countries (Kutuk, 2022). These pilots consistently report benefits such as reduced operating costs (MIT AgeLab, 2021), reliable and continuous service, and lower energy consumption and emissions (Engesser et al., 2023). At the same time, field experience highlights ongoing challenges related to battery range, charging downtime, communication infrastructure, and the need to clearly define each vehicle's Operational Design Domain (ODD), which specifies the physical, environmental, and infrastructural conditions under which autonomous driving can be safely performed (Society of Automotive Engineers (SAE), 2018; Hawkins, Osborne, Parsons, Nicholson, McDermid and Habli, 2022).

The ODD specifies the roadway, environmental, and infrastructure conditions under which an autonomous vehicle can safely operate (National Highway Traffic Safety Administration, 2017; International Organization for Standardization, 2025). These conditions include geometric attributes such as curvature, grade, and lane count, roadway classifications and speed limits, static infrastructure features such as tunnels or railway adjacency, and broader environmental factors such as elevation and lighting. Together, these characteristics define the boundaries of safe automated operation. When applied across a road network, they induce a practical geofence that restricts where the vehicle may travel based on the suitability of local operating conditions.

Given these geofenced operating limits, autonomous vehicles must remain within districts whose roadway and environmental conditions satisfy their ODD requirements. Designing the service region partitioning as a set of contiguous districts allows the operator to allocate vehicles efficiently, maintain balanced coverage, and manage travel distance, responsiveness, and charging needs. Prior work has examined spatial partitioning and region design under various modeling assumptions (Carlsson and Delage, 2013; Carlsson, Behroozi and Mihic, 2018; Carlsson, Liu, Salari and Yu, 2024), and our approach extends this line of research by explicitly embedding various AV-related aspects such as ODD features and dispatching frequency into the partitioning process. However, even a carefully designed partition can degrade when demand shifts across space or time, motivating the need for a framework that accounts for uncertainty in future demand realizations.

In practice, spatial demand is never known precisely, because empirical data are affected by out-of-sample variability and finite-sample noise (Pavone, Treleven and Frazzoli, 2010; Menon and Lee, 2017; Bao, Li, Yang and Gao, 2024). Demand patterns also evolve across seasons, behavioral adjustments, and broader temporal cycles within each annual design horizon (Stroh, Erera and Toriello, 2022; Banerjee, Erera and Toriello, 2022). These two sources of uncertainty, arising from statistical estimation error and from genuine spatial drift, introduce ambiguity in the true distribution of future demands. When this ambiguity is ignored, the resulting design may produce distorted district partitions, misallocation of AVs, and ultimately degraded service quality and operational performance. A Distributionally Robust Optimization (DRO) framework of service design must be employed to incorporate distributional ambiguity explicitly, rather than relying on the assumption that past demand patterns will remain stable.

1.1. Literature Review

We position our work within the broader literature by reviewing prior studies on last-mile services, service region partitioning, and autonomous vehicles along with ODD constraints.

1.1.1. Last-Mile Services

Last-mile services bridge users from one center to their destinations through flexible shuttles, microtransit vans, or rideshare systems. In these operations, vehicles are typically dispatched from a center to serve demands as they arise. Prior research has examined flexible routing strategies, such as hub-and-shuttle systems that integrate fixed-route transit with on-demand feeders (Maheo, Kilby and Van Hentenryck, 2019) and ridesharing-enhanced shuttle operations (Aquad-Perez and Van Hentenryck, 2022). Other studies optimize spatial partitions of the service region (Carlsson et al., 2024), where district-based designs derive optimal service zones for fixed fleets to reduce passenger wait times in transit settings (Li and Quadrifoglio, 2011) and to improve workload balance and travel efficiency (Banerjee et al., 2022; Carlsson et al., 2024). However, these works generally assume known or stable demand patterns. In contrast, our model incorporates AV-specific constraints and operating costs while explicitly addressing ambiguity in the spatial distribution of future demand.

1.1.2. Service Region Partition

Dividing a service region into districts is a classical tactic in transportation design that enables dedicated vehicle assignments. Li and Quadrifoglio (2011) use continuous approximation to design rectangular feeder transit zones for a fixed fleet under a uniform demand density and exogenous dispatch rules. Robust partitioning work by Carlsson and Delage (2013); Carlsson et al. (2018) and the same day delivery model of Banerjee et al. (2022) construct vehicle zones through (weighted) Voronoi schemes around predetermined seed points, assuming known demand distributions and conventional vehicles. Miao, He, Pepin, Han, Hendawi, Khalefa, Stankovic and Pappas (2021) combine quad tree region partitions with data driven distributionally robust vehicle balancing, but the spatial partition is imposed by the quad tree and service quality is measured via imbalance rather than direct user waiting. In contrast, our framework endogenously optimizes the depot location, continuous district boundaries, fleet sizes, and dispatch schedules for AV

fleets, does not rely on Voronoi reference points, and embeds Wasserstein demand ambiguity directly into the cost so that region design and service frequency remain robust to ambiguous spatial demand.

1.1.3. Autonomous Vehicles

AV research emphasizes that the ODD determines the conditions under which AVs can safely operate. Empirical studies show how roadway geometry, lighting, and pedestrian interactions shape AV and delivery-robot performance (Engesser et al., 2023). Broader assessments of AV readiness highlight that ODD boundaries protect systems from the unbounded range of real environments (Fagnant and Kockelman, 2015). Safety-assurance frameworks also note that performance degradations often arise near the edges of these ODDs, where sensing and prediction face the most challenging conditions (Hawkins et al., 2022). Although this literature defines and characterizes ODDs, it does not quantify ODD difficulty or incorporate ODD considerations into operational design for last-mile systems. Existing partition and fleet-planning models implicitly assume conventional vehicles and overlook autonomy-specific environmental constraints. Feature-based ODD modeling provides a more structured perspective. Wittmann, Wang and Lienkamp (2015) formalize ODDs through a high-dimensional specification space of roadway and environmental attributes. Miao, Jia, Li, Lin and Wu (2020) consider mobility-on-demand systems operating within fixed ODD-compatible regions. These works map environmental features to feasibility, but feasibility is treated as a binary condition and does not reflect how the continuous operating difficulty varies across heterogeneous environments. Our approach extends this literature by mapping ODD features to a continuous modular cost function that captures the adaptation effort required for different environmental conditions. This allows continuous ODD complexity to influence operational decisions within a unified optimization framework.

1.2. Contributions

This paper makes several modeling and conceptual as well as solution method contributions to the design of autonomous last-mile transit systems.

1.2.1. Modeling

The primary modeling contribution of this paper is a flexible modeling framework that brings together several core design decisions for autonomous last-mile transit systems. Region partition, depot placement, fleet sizing, and dispatch frequency are typically studied separately because most models do not accommodate these decisions within a unified structure. Our formulation is designed to support this joint consideration. It treats these elements as interdependent rather than isolated components, which allows the model to capture the trade-offs that arise when operational performance and the physical environment must be coordinated.

To make such flexibility possible under the AV setting, we develop a cost-based representation of the ODD features. To this end, prior ODD studies describe environmental and infrastructural attributes or identify regions where automated driving is feasible, but they do not quantify how operating difficulty varies across space. We extend these ideas by constructing a modular cost function that maps multidimensional ODD features to measurable adaptation effort. This representation allows ODD complexity to influence the various design decisions such as the shape of service districts, the suitability of depot locations, and the assignment of fleet resources. It links AV capability considerations directly to the design of the service, reflecting the flexible structure of the overall formulation.

Building on such structural flexibility, we can incorporate more expressive uncertainty modeling of spatial demand patterns through a distributionally robust approach. In reality, last-mile systems face nonstationary demand that is difficult to capture using empirical data alone, and traditional partition models often assume that the true distribution is accessible and stationarily known (uniform or well-defined with closed-form). Our formulation has the capability to treat the distribution as ambiguous within a neighborhood around historical observations and evaluates partitions according to their performance under plausible deviations. This perspective complements the ODD-based cost representation, for environmental difficulty and demand uncertainty can jointly determine which district structures are resilient. The interaction of these two elements yields designs that differ meaningfully from those produced under nominal assumptions and provides insight into how infrastructure conditions and uncertain demand shape operations.

The resulting framework is flexible and general. The modular ODD representation can incorporate additional environmental features, alternative AV capability profiles, or new roadway attributes without changing the form of the cost function. The robust demand component can adopt different ambiguity sets or distributional models, and the partition structure naturally extends to settings with multiple depots, finer or coarser geographic units, or additional operational constraints such as coverage requirements or service-level guarantees. These forms of extensibility

underscore the broader applicability of the model and we return to this theme in the conclusion when discussing potential extensions to other autonomy-aware planning problems.

1.2.2. Solution Method

Our solution method builds on a set of techniques that support the flexibility and generality of the modeling framework. The very first component is the construction of a feature-based ODD dataset from high-resolution geospatial and roadway information. These environmental features are aggregated to the district level, which provides a scalable representation that can accommodate any ODD features or capability profiles without altering the optimization structure.

Despite the flexibility of ODD modeling, the primary computational difficulty emerges when incorporating distributional robustness to model ambiguous demand patterns. DRO problems are demanding because they require optimization over the infinite dimensional probability space, and the situation is further complicated when our objective is nonlinear in the probability density. In our formulation, the service cost depends on square-root density integrals arising from continuous routing approximations, which places the problem outside the range of classical DRO reformulations. The district-level worst-case distribution also interacts with the district boundaries, because any change in the partition modifies the effective support of the demand distribution. As a result, neither the objective nor the feasible region satisfies standard assumptions from the DRO literature. To overcome these obstacles, we derive tractable district-level worst-case distributions that exploit the spatial structure of the routing cost and that can be solved using commonly available convex optimization tools. This approach keeps the robust component computationally manageable. A computational time comparison with the method in Carlsson et al. (2018) is shown in Appendix B.

A further difficulty is the coupling of the district partition with depot placement, fleet sizing, and dispatch frequency. These decisions are rarely optimized together because continuous partitions are infinite-dimensional objects, and their interaction with operational variables produces a design space that is challenging to explore (Carlsson, Lim, Liu, Yu, Arntong and Tan, 2025). In our setting this difficulty is compounded by the nonlinear DRO objective, which further entangles spatial and operational decisions. We address these challenges through two discretizations. The first discretizes the distribution density, which reduces the inner worst-case distribution to a piecewise uniform approximation and yields vanishing approximation error as the discretization is refined. The second discretizes the geographic partition, which converts the continuous region into a tractable combinatorial structure that can be searched efficiently using established heuristic methods. Together, these discretizations enable the DRO inner problem to be solved using standard convex optimization tools and allow the joint partition–operations problem to be addressed at realistic urban scales.

Together, these techniques (summarized via Figure 1) make it possible to solve a problem that would otherwise be analytically intractable. By addressing the difficulties created by ODD spatial heterogeneity, ambiguous demand, and the coupling of multiple operational decisions, the solution approach supports the flexibility and generality of the modeling framework. The same techniques can accommodate richer ODD feature sets, alternative ambiguity models, and additional operational constraints, and we return to these extensions in Section 6.

1.3. Paper Organization

This remaining of this paper is organized as following: Section 2 presents the problem setting of the AV-based last-mile service and outlines the key modeling assumptions. Section 3 develops our partition and service design model, integrating ODD constraints and demand ambiguity. Section 4 describes the discretization approach and solution algorithms used to solve the resulting optimization problems. Section 5 evaluates the framework through a real-world last-mile service case study and compares its performance with existing fixed-route operations. Section 6 concludes the paper and discusses implications for AV-enabled mobility systems.

2. Problem Setting and Assumptions

We study emerging last-mile services operated by AVs, such as pilot programs developed by TIER IV and local mobility operators. In practice, a single depot (e.g. Waymo's depot in San Francisco (Herger, 2025)) serves as the operational base for charging, maintenance, and software updates to achieve economy of scale for fleets on a large region. After partitioning the service region into districts, each district is assigned a dedicated fleet whose AVs reside at the depot between dispatches and is dispatched at scheduled times. The key operational decisions include the depot location, the partition, the fleet sizes, and the dispatch schedules. Designing these components requires balancing

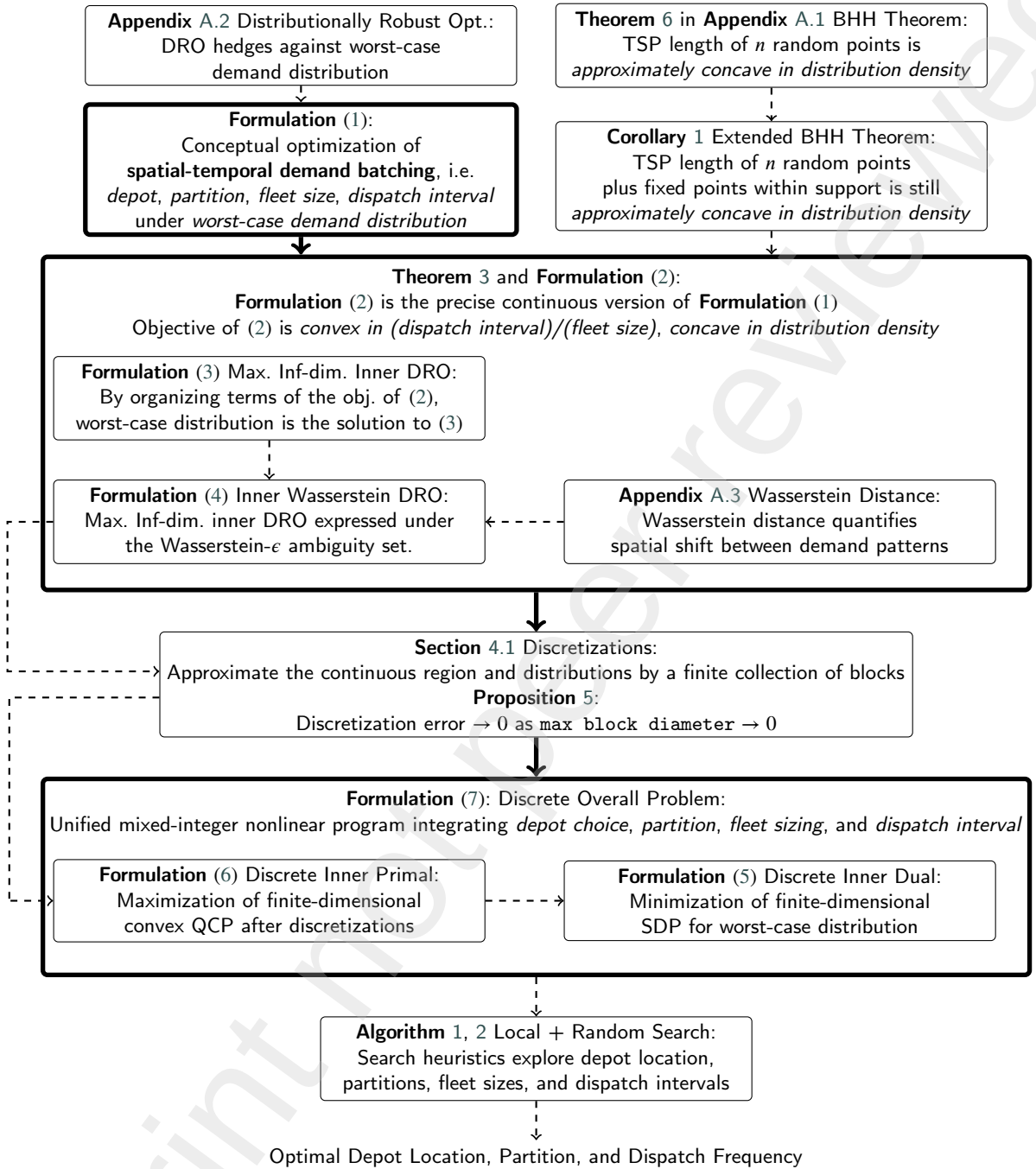


Figure 1: A generalizable solution method for spatial-temporal demand batching problems. Thick boxes represent different reformulations of the overall optimization problem: from conceptual formulation (1), to the full formulation (2), to discretized reformulation (7). Thin boxes summarize the detailed results. For instance, formulation (7) has a tractable semidefinite inner problem (5), and is an exact reformulation of (2) (up to discretization error in Proposition 5). Thick arrows can be interpreted as “reformulated as”. Dashed arrows can be interpreted as “enables”.

the ODD constraints that shape feasible service coverage, the workload distribution across districts, and the demand ambiguity that affects fleet utilization and routing performance.

In our framework, ODD considerations are modeled through spatial features assigned to each location in the service region. These features capture roadway geometry, road type, static infrastructure, environmental context, and interaction complexity, which together determine the operating feasibility of an AV. Each AV type can operate only within districts where all locations' feature values fall inside its admissible range. Therefore, when forming a district and deploying vehicles, the fleet's capabilities must match the district's aggregated ODD features. Since location-level ODD features are fixed, feasibility depends on how locations are grouped into districts (i.e. partition) and on how fleet capabilities are constructed (i.e. training AVs). This embeds ODD heterogeneity into the partition design and links spatial feasibility with vehicle training costs. Figure 2 summarizes the ODD feature taxonomy used in our model.

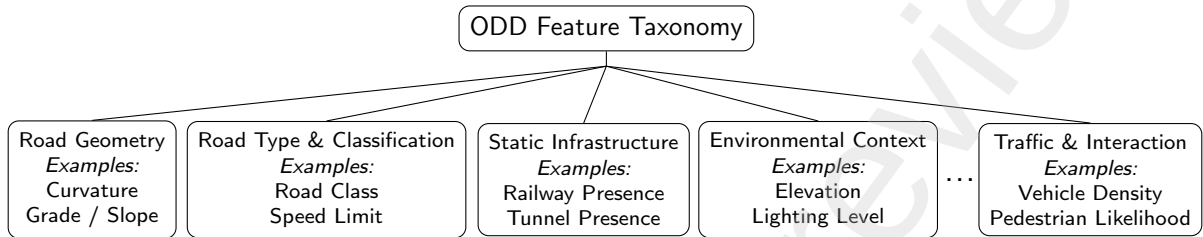


Figure 2: Taxonomy of ODD Feature Categories

Evidence from existing last-mile services shows spatial fluctuations in demand patterns. Commuting demand peaks in residential areas during the morning, commercial demand peaks in business districts in the evening, and seasonal variation shifts demand throughout the year. Fixed service districts combined with static fleet allocations often result in uneven workloads across districts. These imbalances highlight the need for a more robust design.

Given these operational realities, our research question arises: *how should an AV-based last-mile operator jointly optimize the depot location, the service-region partition, the fleet allocation, and the dispatch frequency when the spatial distribution of requests is uncertain and the design must respect ODD-induced feasibility constraints across the region?* We address this question through a Wasserstein distributionally robust design framework coupled with ODD-driven AV cost modeling.

To clarify the modeling scope, we make several structured assumptions regarding ODD cost, decision frequency and uncertainty:

- **ODD feature and cost.** We summarize the ODD difficulty of each district using the elementwise maximum of its location-level ODD features, reflecting the need for the fleet to handle the most challenging conditions within that district. This district summary is then mapped to a scalar ODD cost through a nondecreasing modular function that captures how operational difficulty increases with feature severity.
- **Decision horizon.** Strategic decisions (depot location, district partition) are fixed for one operating year. At the start of each year, the operator updates these decisions using the previous year's data to forecast and re-estimate demand patterns.
- **Temporal scope of the min-max problem.** The min-max objective minimizes the worst-case performance within the upcoming year, not across an indefinite multi-year horizon. This annual framing prevents the model from being dominated by extremely rare events that might occur over a decade, while still capturing meaningful operational variability across days and seasons.
- **Type of ambiguity.** We assume although the true spatial demand distributions are unknown, they are bounded within a Wasserstein ambiguity set centered on the empirical distribution. Each dispatch represents a new realization of spatial demand drawn from this uncertain distribution. The total number of users per dispatch is bounded by a "safe" upper limit (derived from historical peaks), ensuring tractability and robustness. In our case study (Section 5), we assume this ambiguity arises from the intra-day changes of demand patterns, e.g., morning vs evening. But this particular interpretation is not necessary for the general application of our model.

These assumptions define the model's operational logic: depot, and partition decisions are updated annually, while dispatch schedules and fleet sizes adjustable within the year. Ambiguity captures spatial variability in demand, and robustness ensures stable performance against the most adverse yet plausible spatial patterns.

3. Spatial-temporal Demand Batching Models: Partition of Service Region and Demand Arrivals for Last-Mile Autonomous Service Under Uncertainty

This section formulates the distributionally robust optimization model for designing reliable last-mile AV services under spatial demand uncertainty. The model jointly determines the depot location, regional partition, fleet allocation, and dispatch scheduling, integrating operational, spatial, and user considerations. It captures both stochastic demand variations and ODD-related spatial constraints to ensure balanced service performance across all districts even under demand uncertainty.

3.1. Design Decisions and Constraints – Depot Location, Partition, Vehicles, and Dispatch

We present our summary formulation as (1), and unpack each objective function term and constraint in this section.

$$\min_{D, P_{1:k}, V_{1:k}, T_{1:k}} \max_{f \in \mathcal{F}} \underbrace{\text{SYST}(D, P_{1:k}, V_{1:k})}_{\text{provider cost (system)}} + \left(\max_{i=1, \dots, k} \underbrace{\text{PROV}(D, P_i, V_i, T_i, f)}_{\text{provider cost (district)}} + \underbrace{\text{USER}(D, P_i, V_i, T_i, f)}_{\text{user cost}} \right) \quad (1a)$$

$$\text{s.t.} \quad \text{Depot is in the service region,} \quad (1b)$$

$$\text{The whole service region } \mathcal{S} \text{ is covered,} \quad (1c)$$

$$\text{Districts are disjoint,} \quad (1d)$$

$$\text{Each district is closed and contiguous,} \quad (1e)$$

$$\mathbb{P}_f(\text{Workload per AV dispatch is bounded by its capacity}) \geq 1 - \varepsilon, \quad (1f)$$

$$\mathbb{P}_f(\text{Each AV returns to the depot before its next dispatch}) \geq 1 - \varepsilon. \quad (1g)$$

The entire compact service region is denoted by $\mathcal{S}$, and the decision maker selects a single depot location D as a design variable. It provides essential functions such as AV charging, parking, maintenance, and software updates. It serves as the common origin and return point for all vehicles across districts. Without loss of generality, we assume D lies within the service region, that is, $D \in \mathcal{S}$ (1b).

Spatial batching of demand: The decision maker partitions the service region $\mathcal{S}$ into k disjoint districts $P_1, \dots, P_k$ that collectively cover the entire region. This requires exhaustive coverage $\bigcup_{i=1}^k P_i = \mathcal{S}$ (1c), pairwise disjointness almost everywhere $|P_i \cap P_j| = 0$ for all $i \neq j$ (1d), and that each district is both closed and spatially contiguous (1e). Note that since each P_i is closed, boundary points may belong to multiple districts, but their total Lebesgue measure is zero and hence is almost nowhere for an absolute continuous measure.

The decision maker assigns V_i vehicles to serve district P_i , each with capacity Q . These V_i AVs operate in cycles, repeatedly serving demands that accumulate over time within the district. Each AV is dispatched at regular intervals according to a schedule tied to the district's master dispatch interval, as defined in the following dispatch decisions.

Temporal batching of demand: On each district P_i , each of the V_i AVs operates on a periodic dispatch interval of length T_i (maximum time a AV on service). More specifically, the decision maker divides T_i into V_i equal, non-overlapping subintervals of length $C_i := T_i/V_i$ (we will use C_i and T_i/V_i interchangeably), and assign exactly one AV to each subinterval. At the scheduled time marking the end of its assigned subinterval, each AV departs to serve all demands that have accumulated during that subinterval. As a result, the departure order of vehicles within each interval T_i follows the order of their assigned subintervals. Since this sequence repeats in every dispatch interval, each AV effectively operates on a clock of period T_i , departing once every full interval. From each AV's perspective, this induces a "wait-dispatch-clear-return" cycle: each AV first waits at the depot while demands accumulate, then departs at its scheduled dispatch time determined by C_i , traverses the assigned demand locations, and finally returns to the depot, where it remains waiting or refueling until its next dispatch time.

This evenly spaced dispatch structure provides a stable schedule that is independent of subsequent demand uncertainty and offers a practical way to control user waits without increasing AV travel lengths. Keeping other elements fixed, such as the fleet size, district partition, and the set of users assigned to each AV, evenly dividing T_i across the V_i vehicles minimizes the longest gap between departures and therefore helps reduce the maximum user wait at depot. If we compare this with a VRP-style batch-dispatch scheme in which all AVs depart at the end of the interval but assignments are not reoptimized, then each AV would still serve the same users in both regimes, and its routing distance would be unchanged; only the user waits would be larger under the batched schedule. VRP-based approaches could

in principle further reduce travel distance by reoptimizing user–vehicle assignments each interval, but this requires solving a large matching or routing problem in real time, which may be operationally difficult. Figure 3 illustrates this rolling-wave dispatch scheme.

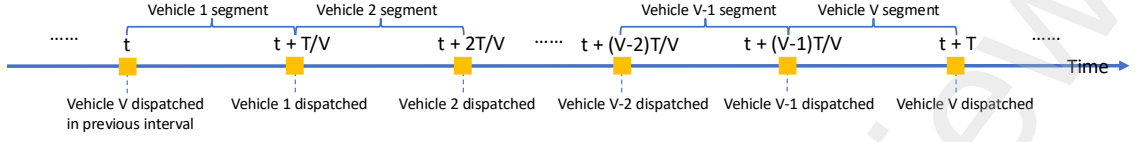


Figure 3: Dispatches of Vehicles per Interval on a Single District

To ensure operational feasibility under uncertain demand realizations, the fleet sizing and dispatch decisions are subject to two chance constraints. The first constraint (1f) requires that the total number of demands assigned to a single AV in district P_i during one dispatch does not exceed its capacity Q with probability at least $1 - \varepsilon$ under distribution f . Denoting the random number of demands per dispatch as N_i , (1f) can be written as $\mathbb{P}_f(N_i \leq Q) \geq 1 - \varepsilon$. The second constraint (1g) ensures that every AV completes its assigned service and returns to the depot before its next scheduled dispatch, also with probability at least $1 - \varepsilon$: $\mathbb{P}_f(\text{service time per dispatch} \leq T_i) \geq 1 - \varepsilon$. We impose these constraints probabilistically rather than as hard constraints because commonly used temporal demand models such as Poisson processes, have strictly positive probability of having arbitrarily large number of arrivals within any fixed duration. In these cases, such hard constraints would be infeasible unless fleets were uncapacitated. By instead allowing a small violation probability, the chance constraints ensure high-probability feasibility for capacity and timing without forcing excessive overprovisioning.

3.2. Cost Objectives

We evaluate the system performance by combining provider and user costs, which together capture both operational efficiency from the provider's perspective and service quality from the users' view. The provider cost consists of the system-level and the district-level costs, while the user cost only includes the district-level cost. The overall objective is to minimize the system-level provider cost plus the maximum district-level cost to ensure fairness and balanced performance across all districts.

3.2.1. System-Level Provider Cost

First, the system-level provider cost depends on the placement of the depot itself, which incurs location-specific expenses. The depot cost, denoted by $\phi(D)$, captures location-specific expenses such as rent, accessibility, or infrastructure associated with placing the depot at location D . The function ϕ can be treated as a model parameter, either derived from external geographic or economic data or selected to reflect policy or planning priorities.

Besides, we account for the fixed investment in AV deployment, represented by the total number of AVs assigned across all districts. Let l denote the unit cost of each AV. The total fleet cost is therefore $l \sum_{i=1}^k V_i$, representing the fixed investment associated with the number of AVs assigned across all districts.

We also include the operational design domain cost from AV operations across districts. The ODD cost captures the environmental and contextual complexity that AVs may encounter while operating within a district. Originating from National Highway Traffic Safety Administration (2017) and widely adopted in autonomous vehicle safety and deployment studies (Koopman, Fratrik et al., 2019; Lee, Nayeer, Garcia, Agrawal and Liu, 2020; Hawkins et al., 2022; Committee, 2021), this cost accounts for physical and infrastructural features such as road geometry (e.g., curvature, grade), speed limit zones, presence of intersections, type of lane markings, and proximity to fixed infrastructure (e.g., rail crossings or pedestrian-dense areas). These contextual factors influence both the feasibility and operational efficiency of AVs.

We model the ODD cost in a modular manner, reflecting that AV capabilities can be incrementally enhanced to handle additional environmental complexities. Specifically, we represent contextual conditions using a bounded feature mapping $\Omega : S \rightarrow \mathbb{R}^{d^{\text{odd}}}$, where $\Omega(\mathbf{x})$ is a d^{odd} -dimensional vector of ODD features at location $\mathbf{x} \in S$. Each feature dimension is defined such that larger values indicate greater operational difficulty. To summarize the contextual challenge within district P_i , we define a district summary vector ω_i as the elementwise maximum of $\Omega(\mathbf{x})$ across all

$\mathbf{x} \in P_i$, i.e.,

$$(\boldsymbol{\omega}(P_i))_j := \sup_{\mathbf{x} \in P_i} (\boldsymbol{\Omega}(\mathbf{x}))_j, \quad \forall i = 1, \dots, k; j = 1, \dots, d^{\text{odd}}.$$

This construction captures the most challenging conditions present within each district, ensuring that the planning remains safety-first, and accounts for the hardest case each AV may face.

The nondecreasing modular cost function $\mathcal{J}(\boldsymbol{\omega}(P_i))$ then maps the district summary vector $\boldsymbol{\omega}(P_i)$ to a scalar ODD cost, quantifying how the overall operational challenge increases with contextual difficulty. We adopt an affine modular specification of the form

$$\mathcal{J}(\boldsymbol{\omega}(P_i)) = \eta_0 + \sum_{j=1}^{d^{\text{odd}}} \eta_j \cdot (\boldsymbol{\omega}(P_i))_j,$$

where η_0 represents a base cost associated with minimal operational requirements for the whole region $\mathcal{S}$, and each coefficient $\eta_j \geq 0$ captures the marginal adaptation cost attributable to the j th ODD feature, such as curvature, intersection density, or pedestrian exposure. This affine modular representation reflects the additive nature of AV capability enhancement, where individual functional modules can be upgraded independently according to environmental complexity. In the case that feature j and feature j' are not “independent” – e.g., if j is related to visibility, and j' is related to road condition, they may both be influenced by precipitation – this additive assumption is still sound given the redundancy mindset in the AV community and companies. Both the feature mapping $\boldsymbol{\Omega}$ and the cost parameters $\{\eta_j\}$ can be estimated from domain knowledge, simulation-based evaluation, or empirical deployment data.

The ODD cost on each district only depends on the district topology P_i (and does not scale with the number of AVs), as it reflects the one-time expense of enabling the required capability modules for operating within that district. These modules are installed or activated based on the most challenging conditions present in P_i , and once configured, can support any number of vehicles dispatched within the district. Let $F_i := F(P_i)$ denote the ODD cost on P_i , then

$$F(P_i) = \mathcal{J}(\boldsymbol{\omega}(P_i)) = \eta_0 + \sum_{j=1}^{d^{\text{odd}}} \eta_j \cdot (\boldsymbol{\omega}(P_i))_j.$$

Since the base cost η_0 is common to all districts and does not influence the relative comparison among designs, it can be excluded from the objective. Without loss of generality, the district-level ODD cost is therefore

$$F(P_i) = \sum_{j=1}^{d^{\text{odd}}} \eta_j \cdot (\boldsymbol{\omega}(P_i))_j.$$

We evaluate ODD complexity at the system level by considering the most demanding district, since the maximum ODD cost determines the level of capability that must be supported fleetwide. Accordingly, we denote the maximum ODD cost across all districts by

$$O(P_{1:k}) := \max_{i=1, \dots, k} F(P_i),$$

which enters the overall objective as the ODD component of the system cost.

Our modeling approach builds on and extends existing feature-based ODD modeling frameworks in the AV literature. Prior work represents the ODD through structured environmental attributes, either by defining a high-dimensional specification space that quantifies road, traffic, and environmental characteristics for identifying system boundaries Wittmann et al. (2015), or by characterizing ODD-compatible geographic regions when modeling mobility-on-demand operations Miao et al. (2020). These formulations share our premise that ODD can be decomposed into measurable features. However, they ultimately use these features to determine whether an AV may operate, rather than how costly it is to operate across heterogeneous environments. In contrast, we map ODD features to a continuous modular cost function $\mathcal{J}(\boldsymbol{\omega})$, quantifying the marginal adaptation effort associated with each environmental dimension. This shifts ODD from a binary feasibility condition to a decomposable cost driver and enables explicit trade-offs among district design, uncertainty robustness, and AV capability adaptation.

Consequently, we have the system-level provider cost as

$$\text{SYST}(D, P_{1:k}, V_{1:k}) = \phi(D) + l \sum_{i=1}^k V_i + O(P_{1:k}).$$

3.2.2. District-Level Provider Cost

The district-level provider cost on district P_i consists of two dispatch-driven components, namely the *in-district travel cost* and the *linehaul cost*, both resulting from routing vehicles to serve accumulated demands. We first discuss the district-level operations which enables us to formulate the dispatch-related travel costs.

Given the demand location information available at the time of each dispatch, the ideal route is a traveling salesman (TSP) tour that includes the depot and all demand locations for minimal travel length. To obtain the corresponding TSP length, it is a common practice to use the classical BHH theorem (Theorem 6) (Beardwood, Halton and Hammersley, 1959) as a continuous approximation of the minimal TSP travel distance (Carlsson and Delage, 2013; Carlsson et al., 2018; Banerjee et al., 2022; Carlsson et al., 2024). More specifically, Theorem 6 suggests that if n demand locations are randomly sampled from the density f^{P_i} over district P_i , then the TSP travel distance through the n locations is asymptotically given by $\beta\sqrt{n} \int_{P_i} \sqrt{f^{P_i}}$, where we use the notation f^{P_i} as the conditional distribution density of demand locations over district P_i . This convergence holds with consistency in the sense that the scaled travel length $L_n/\sqrt{n}$ converges in probability to the limiting expression.

In addition, in our formulation all AVs also need to travel through the depot per dispatch, and the depot location D might not be in the district P_i , causing an additional distance bias larger than $2 \min_{\mathbf{x} \in P_i} d(\mathbf{x}, D)$ on the BHH TSP approximation, where d is the metric used for distance. To mitigate this type of approximation bias, we follow a routing policy similar to that in Banerjee et al. (2022), and decompose the travel of each dispatch into two components: the travel within the district (in-district TSP travel) and the travel between the depot and the district (linehaul travel). Specifically, we assume that each AV is first routed from the depot to the closest location in its assigned district. This location may coincide with the depot itself if the depot lies within the district. The AV then completes a TSP tour that starts at this closest location and passes through all assigned demand locations of that dispatch within the district, which provides an upper bound on the minimal travel length. We refer to the closest location of each district from the depot as the “root” $\mathbf{x}_0^i$ of that district.

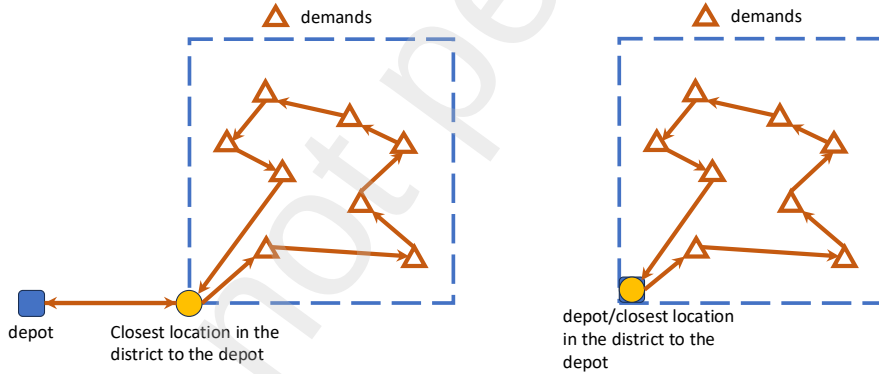


Figure 4: Single-district Single-vehicle Route per Dispatch

The depot outside the district (*left*): Each vehicle dispatches from the depot. It first travels to the root of its district, then starts its TSP tour, and comes back to the root, and finally travels back to the depot. The depot inside the district (*right*): Each vehicle dispatches from the depot, directly starts its TSP tour, and comes back to the depot.

For the in-district TSP travel, the root $\mathbf{x}_0^i$ in P_i is not random but determined by the depot D and the district shape P_i , so the classical BHH theorem (Beardwood et al., 1959) (Theorem 6) does not immediately apply. To address this, we propose a generalized version of BHH theorem as Corollary 1.

Corollary 1 (Consistent Approximation with One Root). *Consider the probability measure $\mathbb{P}$ over its support P , and let f be the probability density of the absolutely continuous part of $\mathbb{P}$. Let one fixed point $\mathbf{x}_0 \in P$, and $\mathbf{X}_1, \dots, \mathbf{X}_n, n \geq 1$, be a set of i.i.d. random locations following $\mathbb{P}$. Then the length L_n^+ of a shortest TSP tour through the collection of points $\{\mathbf{x}_0, \mathbf{X}_1, \dots, \mathbf{X}_n\}$ still satisfies*

$$\frac{L_n^+}{\sqrt{n}} \rightarrow \beta \int_P \sqrt{f(\mathbf{x})} d\mathbf{x}$$

with probability 1 as $n \rightarrow \infty$, where β is a constant only depending on the metric space. When the metric space is $(\mathbb{R}^2, d^{Euc})$, $\beta \approx 0.7120$.

By Corollary 1, the in-district travel cost (distance) on P_i can be consistently approximated by $\beta\sqrt{n} \int_{P_i} \sqrt{f^{P_i}}$, where f^{P_i} is the conditional density on P_i and $\beta \approx 0.7120$ if we consider the Euclidean space. The corollary relies on a rooted version of the BHH theorem, which shows that adding a single fixed point inside P_i to traverse by a TSP tour does not change the $\sqrt{n}$ asymptotic scaling of the TSP length. As a result, this approximation captures the leading $O(\sqrt{n})$ term of the in-district travel distance, which reflects the spatial dispersion of the n demand points without solving the combinatorial TSP for each dispatch. The depot location does not play a role in this approximation, for the in-district travel depends only on the distribution of demand points within P_i . The linehaul cost of traveling between the depot and the district is therefore modeled separately.

Let K_i denote the fixed linehaul travel cost (distance) from the depot D to the root in district P_i , which serves as the AVs' entry and exit point for every dispatch. This cost is incurred regardless of the demand realization, as every dispatch must travel between D and P_i . The fixed linehaul distance K_i depends on the depot location D and the geometry of P_i .

Since both in-district TSP and linehaul travel distance scale with the number of dispatches, which also depends on the dispatch interval T_i in a unit time period, we need amortized cost to standardize the time scope. As a result the in-district provider cost is

$$\text{PROV}(D, P_i, V_i, T_i, f) = \frac{V_i}{T_i} \left(K_i + \beta \sqrt{\Lambda^{P_i} \frac{T_i}{V_i}} \int_{P_i} \sqrt{f^{P_i}} \right) = \frac{K_i}{T_i/V_i} + \beta \sqrt{\frac{\Lambda^{P_i}}{T_i/V_i}} \int_{P_i} \sqrt{f^{P_i}},$$

where Λ^{P_i} is the arrival rate over district P_i , whose explicit definition will be provided in the next subsection.

3.2.3. District-Level User Cost

User cost consists of two components: the longest wait a user may experience before being picked up (maximum user wait), and the longest time a user may spend on board the AV before arrival (maximum user travel). We model both components explicitly and convert them to cost (evaluated by amortized AVs' travel distance) using a weighting coefficient w_r , which reflects the decision maker's emphasis on user satisfaction relative to operational efficiency. We adopt the maximum cost to ensure fairness, by safeguarding against extreme delays or long travel times for any individual user, especially those in the most disadvantaged locations within a district. In the case study Section 5, we illustrate how the transit provider may want to explore a range of w_r values to compute operational metrics for nonprofit organization reporting purposes.

The longest wait a user may experience on district P_i is $C_i = T_i/V_i$, which occurs when the demand arrives exactly at the start of a dispatch subinterval. This captures the maximum possible delay before a AV becomes available to board the user. Hence, the maximum user wait cost is $w_r C_i$.

The longest on-board travel time for a user in district P_i consists of half the linehaul travel from the depot to the district boundary, plus the full in-district routing distance. This results in a total on-board travel distance of $K_i/2 + \beta\sqrt{n} \int_{P_i} \sqrt{f^{P_i}}$, where K_i denotes the round-trip depot-to-district distance, and the second term reflects in-district route distance under the conditional demand distribution f^{P_i} . Hence, the maximum user travel cost is

$$\text{USER}(D, P_i, V_i, T_i, f) = w_r \left(\frac{1}{s} \left(\frac{K_i}{2} + \beta \sqrt{\Lambda^{P_i} \frac{T_i}{V_i}} \int_{P_i} \sqrt{f^{P_i}} \right) + T_i/V_i \right),$$

where s is assumed to be the constant speed of AVs.

Combining the provider's and users' costs, we evaluate the total district-level cost as the sum of operational, spatial, and user-related components defined above. To ensure fairness and service reliability across the entire service region, the overall objective is to minimize the maximum district-level cost among all k districts. This min-max formulation prevents excessive cost concentration in any single district and favors balanced service quality under heterogeneous spatial demand conditions.

3.2.4. Total Cost Balances Provider Cost And Single User Cost

We have thus described the objective function as a weighted sum of the provider cost and a single (worst-case) user's cost. We acknowledge that this is not the only way to formulate an objective function and may not be suitable for all

last-mile transportation services. In our particular case, this objective function is a generalization of what we observed in practice with last-mile transportation providers that have two objectives: minimizing cost and maximizing ridership. For instance, a nonprofit transportation provider needs to report to their board each fiscal year on the metric of “cost per rider”. While the cost depends on the number of vehicles, driving distances, and so on, the denominator (number of riders) depends on each individual rider’s experience, i.e., wait and ride times. Therefore, we find it appropriate to have an objective function that explicitly balances the provider cost and an individual user’s cost. But for other, for-profit businesses that have different objectives, e.g., maximizing profit via optimal service design and fare pricing, one may want to modify the objective function.

3.3. Stochastic Demands Modeling and Ambiguity Set

In our model, we assume the spatially inhomogeneous Poisson arrival of demands (Pavone et al., 2010; Menon and Lee, 2017; Bao et al., 2024) due to its memoryless property. We assume an overall demand arrival rate Λ over the whole region S during a period, and the spatially uncertain future demand is captured by a continuous distribution $\underline{f}$ supported on S , where a higher density $\underline{f}(\mathbf{x})$, $\mathbf{x} \in S$ implies greater probability that one arrived demand appear at $\mathbf{x}$. Then $\Lambda \int_{P_i} \underline{f}(\mathbf{x})$ denotes the arrival rate over district P_i .

Constraints (1f) and (1g) ensure operational feasibility under stochastic demand realizations for any distribution $\underline{f}$. The first constraint guarantees that the workload assigned to each AV does not exceed its capacity with probability at least $1 - \epsilon$, while the second ensures that each AV can complete its route and return to the depot before the next dispatch with the same confidence level. Both conditions are formulated under the spatially inhomogeneous Poisson arrival model and can be transformed into deterministic lower bounds on fleet sizes for tractable optimization.

Proposition 2 (Fleet Size Lower Bound under Chance Constraints). *Under the spatially inhomogeneous Poisson arrival process with overall rate Λ , spatial density $\underline{f}$, vehicle speed s and vehicle capacity Q , the chance constraints (1f) and (1g) imply the following deterministic lower bound on fleet size V_i for district P_i .*

$$V_i \geq \frac{\Lambda T_i}{\min\{\bar{\lambda}(Q, \epsilon), \bar{\lambda}(\bar{Q}_i, \epsilon)\}} \int_{P_i} \underline{f}(\mathbf{x}) d\mathbf{x},$$

where Q denotes the vehicle capacity, and $\bar{\lambda}(N, \epsilon)$ is defined as

$$\bar{\lambda}(N, \epsilon) := \sup \left\{ \lambda \geq 0 : \sum_{k=0}^N \exp(-\lambda) \frac{\lambda^k}{k!} \geq 1 - \epsilon \right\}.$$

Here $\bar{Q}_i$ represents the maximum number of demands that can be served within one dispatch interval without violating the return-time constraint,

$$\bar{Q}_i = \left\lceil \frac{sT_i - K_i}{\beta \int_{P_i} \sqrt{f(\mathbf{x})} d\mathbf{x}} \right\rceil^2.$$

Proof. • $\mathbb{P}_f(N_i \leq Q) \geq 1 - \epsilon$: Recall Q is the vehicle capacity and N_i is the number of demands assigned to a single AV in district P_i during one dispatch. Under a Poisson arrival process on P_i with rate

$$\lambda_i := \frac{\Lambda T_i}{V_i} \int_{P_i} \underline{f}(\mathbf{x}),$$

we have $N_i \sim \text{Poisson}(\lambda_i)$. Since $\mathbb{P}_f(N_i \leq Q)$ is decreasing in λ_i , the chance constraint $\mathbb{P}_f(N_i \leq Q) \geq 1 - \epsilon$ is equivalent to requiring that λ_i does not exceed the largest mean $\bar{\lambda}(Q, \epsilon)$ defined as

$$\bar{\lambda}(Q, \epsilon) := \sup \left\{ \lambda \geq 0 : \sum_{k=0}^Q \exp(-\lambda) \frac{\lambda^k}{k!} \geq 1 - \epsilon \right\}.$$

Thus the constraint becomes $\lambda_i \leq \bar{\lambda}(Q, \epsilon)$, which yields the fleet size lower bound

$$V_i \geq \frac{\Lambda T_i}{\bar{\lambda}(Q, \epsilon)} \int_{P_i} \underline{f}(\mathbf{x}),$$

because $\bar{\lambda}(Q, \epsilon)$ can be precomputed. This constraint links fleet size, dispatch interval, and spatial demand mass, and ensures that the capacity violation probability in district P_i does not exceed ϵ .

- $\mathbb{P}_f(\text{service time per dispatch} \leq T_i) \geq 1 - \epsilon$: Let $L_i(N_i, P_i, f)$ denote the expected estimated length of travel per AV per dispatch. If we assume the average AV speed to be s , then the total travel time per AV per dispatch is $(K_i + L_i(N_i, P_i, f))/s$. As a result, the chance constraint is

$$\mathbb{P}_f \left(\frac{K_i + L_i(N_i, P_i, f)}{s} \leq T_i \right) \geq 1 - \epsilon \Leftrightarrow \mathbb{P}_f (L_i(N_i, P_i, f) \leq sT_i - K_i) \geq 1 - \epsilon.$$

With Corollary 1, we can replace $L_i(N_i, P_i, f)$ and obtain the reformulated chance constraint

$$\mathbb{P}_f (N_i \leq \bar{Q}_i) \geq 1 - \epsilon, \text{ where } \bar{Q}_i := \left(\frac{sT_i - K_i}{\beta \int_{P_i} \sqrt{f(\mathbf{x})}} \right)^2.$$

Similarly, the original chance constraint is equivalent to $\lambda_i \leq \bar{\lambda}(\bar{Q}_i, \epsilon)$, leading to the fleet size lower bound

$$V_i \geq \frac{\Lambda T_i}{\bar{\lambda}(\bar{Q}_i, \epsilon)} \int_{P_i} f(\mathbf{x}).$$

In summary, the chance constraints (1f) and (1g) are equivalent to

$$V_i \geq \frac{\Lambda T_i}{\min \{ \bar{\lambda}(Q, \epsilon), \bar{\lambda}(\bar{Q}_i, \epsilon) \}} \int_{P_i} f(\mathbf{x}).$$

□

In practice, distribution f is estimated from limited and potentially outdated observations. The realized spatial demand field may deviate from some nominal estimate $\hat{f}$ due to sampling noise, temporal shifts in user behavior, or exogenous environmental changes. To address this, we model f as uncertain and consider a collection of plausible densities surrounding $\hat{f}$, thereby capturing ambiguity in the underlying spatial demand process. Both the objective function (1a) and the chance constraints (1f), (1g) are subject to this ambiguity.

We construct the ambiguity set $\mathcal{F}$ using the Wasserstein metric to reflect realistic spatial variations in demand. Specifically, we define

$$\mathcal{F} := \{ f \in L_1(S) : W(f, \hat{f}) \leq \epsilon \},$$

where $W(\cdot, \cdot)$ denotes the Wasserstein distance between two spatial densities and $\epsilon > 0$ specifies the allowable deviation from the nominal $\hat{f}$. This formulation captures the intuition that moderate relocation or dispersion of demand locations should not drastically alter the system design. Under this construction, the robust optimization problem seeks the design variables $(D, P_{1:k}, V_{1:k}, T_{1:k})$ that minimize the worst-case expected cost across all $f \in \mathcal{F}$, thus the min-max structure in (1). This distributionally robust formulation ensures that the obtained design remains reliable under all plausible demand realizations consistent with the observed data and spatial structure.

The ambiguity radius ϵ controls the conservativeness of the design: a larger value implies greater protection against demand ambiguities, while a smaller value yields a more data-driven but less robust solution. In practice, ϵ can be calibrated through cross-validation (Gao, 2023; Hao and Zhang, 2025) by evaluating out-of-sample performance across multiple historical demand snapshots and selecting the smallest value that stabilizes the worst-case cost. Alternatively, when only limited samples are available, statistical guarantees from the DRO literature (Shafieezadeh Abadeh, Nguyen, Kuhn and Mohajerin Esfahani, 2018; Blanchet, Kang and Murthy, 2019; Blanchet, Murthy and Si, 2022; Gao, 2023; Gao, Chen and Kleywegt, 2024) suggest scaling the radius as $\epsilon = \epsilon_0/\sqrt{n}$ for some constant $\epsilon_0 > 0$, where n is the number of observed demand points. This data-dependent adjustment balances robustness and efficiency, ensuring that the true distribution lies within the Wasserstein ball $\mathcal{F}$ with high probability.

3.4. Composing Overall Model

Then we flesh out the details of (1) as (2). Note (2) is a nonlinear, semi-infinite optimization problem with Wasserstein ambiguity and nonlinear dependence on demand density f . Our next result proves that this problem has a reasonably nice, partially convex-concave structure, which is desirable for a min-max problem.

Theorem 3 (Overall Formulation). *Consider the metric space $(\mathbb{R}^2, d)$, where vehicles operate at a constant speed s , and let the Wasserstein ambiguity set have radius ϵ . Then the overall problem is equivalent to formulation (2). The objective function (2a) is **convex** in the decision variables $(K_i, T_i/V_i)$ and **concave** in the uncertain density f , under fixed depot location D and partition $P_{1:k}$.*

$$\min_{D, P_{1:k}, T_{1:k}, V_{1:k}, K_{1:k}, F_{1:k}} \phi(D) + l \sum_{i=1}^k V_i + O + \max_{i=1, \dots, k} \max_{f \in \mathcal{F}} \left[\frac{K_i}{T_i/V_i} + \beta \sqrt{\frac{\Lambda}{T_i/V_i}} \int_{P_i} \sqrt{f} + w_r \left(\frac{1}{s} \left(\frac{K_i}{2} + \beta \sqrt{\Lambda T_i/V_i} \int_{P_i} \sqrt{f} \right) + T_i/V_i \right) \right] \quad (2a)$$

$$s.t. \quad D \in S \quad (2b)$$

$$\cup_{i=1}^k P_i = S \quad (2c)$$

$$|P_i \cap P_j| = 0, \forall i, j : i \neq j \quad (2d)$$

$$P_i \text{ is closed and contiguous}, \forall i = 1, \dots, k \quad (2e)$$

$$\mathcal{F} := \{f \in L_1(S) : W(f, \hat{f}) \leq \epsilon\} \quad (2f)$$

$$K_i = 2 \min_{\mathbf{x} \in P_i} d(\mathbf{x}, D), \forall i = 1, \dots, k \quad (2g)$$

$$\omega_i \geq \Omega(\mathbf{x}), \forall \mathbf{x} \in P_i \quad (2h)$$

$$F_i = \sum_{j=1}^{d_{\text{odd}}} \eta_j \cdot (\omega_i)_j, \forall i = 1, \dots, k \quad (2i)$$

$$O \geq F_i, \forall i = 1, \dots, k \quad (2j)$$

$$V_i \geq \frac{\Lambda T_i}{\bar{\lambda}(Q, \epsilon)} \int_{P_i} f(\mathbf{x}) \quad (2k)$$

$$\bar{Q}_i = \left(\frac{s T_i - K_i}{\beta \int_{P_i} \sqrt{f^{P_i}(\mathbf{x})}} \right)^2 \quad (2l)$$

$$V_i \geq \frac{\Lambda T_i}{\bar{\lambda}(\bar{Q}_i, \epsilon)} \int_{P_i} f(\mathbf{x}) \quad (2m)$$

Proof. By definition, the system-level provider cost is

$$\text{SYST}(D, P_{1:k}, V_{1:k}) = \phi(D) + l \sum_{i=1}^k V_i + O(P_{1:k}).$$

Since the system-level provider cost is independent of the distribution, we can move the maximization over the ambiguity set inside and the remainder of the objective (district-level cost) is

$$\max_{f \in \mathcal{F}} \max_{i=1, \dots, k} \text{PROV}(D, P_i, V_i, T_i, f) + \text{USER}(D, P_i, V_i, T_i, f).$$

The overall ODD cost $O(P_{1:k}) := \max_i F_i$, and F_i is equal to $J(\omega(P_i))$ (2i), where $\omega(P_i)$ is the elementwise maximum of $\Omega(\mathbf{x})$ over $\mathbf{x} \in P_i$, i.e. $\omega(P_i) \geq \Omega(\mathbf{x}), \forall \mathbf{x} \in P_i$ (2h).

Then without loss of generality, we use the amortized travel distance as the unit of the overall cost objective, and convert other units into the equivalent distance. On each district P_i , under the Poisson arrival assumption with distribution f , the conditional arrival rate $\Lambda^{P_i} := \Lambda \int_{P_i} f$, then by Corollary 1 we can obtain the approximated in-district TSP length growth rate for per-unit time as

$$\beta \sqrt{\Lambda^{P_i}} \int_{P_i} \sqrt{f^{P_i}} = \beta \sqrt{\Lambda \int_{P_i} f} \int_{P_i} \sqrt{\frac{f}{\int_{P_i} f}} = \beta \sqrt{\Lambda} \int_{P_i} \sqrt{f}.$$

Then plugging this rate expression into PROV yields

$$\text{PROV}(D, P_i, V_i, T_i, f) := \frac{K_i}{T_i/V_i} + \beta \sqrt{\frac{\Lambda^{P_i}}{T_i/V_i}} \int_{P_i} \sqrt{f^{P_i}} = \frac{K_i}{T_i/V_i} + \beta \sqrt{\frac{\Lambda}{T_i/V_i}} \int_{P_i} \sqrt{f},$$

where K_i is the fixed linehaul cost. As defined previously, K_i is the roundtrip distance from the depot to the root of P_i and can be expressed as $2 \min_{\mathbf{x} \in P_i} d(\mathbf{x}, D)$ (2g).

Similarly plugging the rate expression into USER yields

$$\begin{aligned} \text{USER}(D, P_i, V_i, T_i, f) &:= w_r \left(\frac{1}{s} \left(\frac{K_i}{2} + \beta \sqrt{\Lambda^{P_i} T_i / V_i} \int_{P_i} \sqrt{f^{P_i}} \right) + T_i / V_i \right) \\ &= w_r \left(\frac{1}{s} \left(\frac{K_i}{2} + \beta \sqrt{\Lambda T_i / V_i} \int_{P_i} \sqrt{f} \right) + T_i / V_i \right). \end{aligned}$$

Summing the $\text{PROV}(D, P_i, V_i, T_i, f)$ and $\text{USER}(D, P_i, V_i, T_i, f)$ yields the cost objective.

Finally switching the two maximization operators, imposing constraints (1b)–(1e) as (2b)–(2e), and Wasserstein ambiguity set constraint (2f) yield Theorem 3. Since all decisions are positive and there's no interaction term across districts, the cost objective is convex with respect to (K_i, F_i, C_i) , and concave on f with fixed D and $P_{1:k}$. $\square$

4. Solution Methodology

In this section, we first present tractable discrete approximations for both partition and distribution. We then formulate and discretize the inner problem of (2) to identify the worst-case distribution and establish its approximation error bound. Next, we combine these results into a tractable overall discrete approximation formulation. We propose a tailored search heuristic adapted from Validi, Buchanan and Lykhovyd (2022).

4.1. Discrete Approximations

To turn our continuous-space, continuous-distribution model into a tractable finite-dimensional problem, we approximate both the region partition and the distribution by block-wise discretizations.

We first discretize the service region S into a collection of small, contiguous and closed “blocks” $\{B_j\}_{j=1}^N$. Each block is treated as an atomic unit for assignment, and we reformulate the continuous partition $\cup_{i=1}^k P_i = S$ (2c), $|P_i \cap P_{i'}| = 0, \forall i \neq i'$ (2d) as the following block-assignment problem

$$\begin{aligned} \min_{A_1, \dots, A_k} \quad & \max_{i=1, \dots, k} \text{Cost}(A_i) \\ \text{s.t.} \quad & A_i \cap A_{i'} = \emptyset, \forall i \neq i' \\ & \cup_{i=1}^N A_i = \{1, \dots, N\}, \end{aligned}$$

where $A_i \subset \{1, \dots, N\}$ is the set of blocks assigned to district i , and $\text{Cost}(A_i)$ denotes the worst-case cost on the approximated district $\hat{P}_i := \cup_{j \in A_i} B_j$ using the block-union. By viewing each block as a node in a planar graph (adjacency by shared edges), this becomes an equivalent planar graph-partition problem. As a result, we can capture the node assignment using binary decisions $\{z_{ji}\}_{j=1, \dots, N; i=1, \dots, k}$ that indicate whether block B_j belongs to district $\hat{P}_i$, i.e.,

$$z_{ji} := \begin{cases} 1, & \text{if } B_j \in A_i \\ 0, & \text{otherwise.} \end{cases}$$

Next we approximate any probability distribution $\mathbb{P}$ on S by a block-wise uniform histogram. If $m_j = \mathbb{P}(B_j)$ is the mass over block B_j , we can construct an absolutely continuous density $\tilde{f}$ by the step-function

$$\tilde{f}(\mathbf{x}) = \sum_{j=1}^N \frac{m_j}{|B_j|} \mathbf{1}(\mathbf{x} \in B_j),$$

where $\mathbf{1}(\mathbf{x} \in B_j) = 1$ if $\mathbf{x} \in B_j$, otherwise $\mathbf{1}_{B_j}(\mathbf{x}) = 0$, and $\tilde{f}$ is uniform over each block. This block-wise histogram approximation preserves the structure of the inner maximization (3) because the cost function depends only on integrals of $\sqrt{f}$ over districts, which can be well-approximated by summing over blocks.

4.2. Discrete Worst-Case Distribution Inner Problem

We isolate the inner maximization problem in (2) to understand what the worst-case distribution is. Fixing the design decisions and focusing on a single district P_i , we obtain

$$\begin{aligned} & \max_{f \in \mathcal{F}} \frac{K_i}{T_i/V_i} + \beta \sqrt{\frac{\Lambda}{T_i/V_i}} \int_{P_i} \sqrt{f} + w_r \left(\frac{1}{s} \left(\frac{K_i}{2} + \beta \sqrt{\Lambda T_i/V_i} \int_{P_i} \sqrt{f} \right) + T_i/V_i \right) \\ &= \max_{f \in \mathcal{F}} \frac{K_i}{T_i/V_i} + \left(\frac{1}{\sqrt{T_i/V_i}} + \frac{w_r}{s} \sqrt{T_i/V_i} \right) \beta \sqrt{\Lambda} \int_{P_i} \sqrt{f} + \frac{w_r}{2s} K_i + w_r T_i/V_i \\ &= \frac{K_i}{T_i/V_i} + \left(\frac{1}{\sqrt{T_i/V_i}} + \frac{w_r}{s} \sqrt{T_i/V_i} \right) \left(\max_{f \in \mathcal{F}} \beta \sqrt{\Lambda} \int_{P_i} \sqrt{f} \right) + \frac{w_r}{2s} K_i + w_r T_i/V_i \\ & \text{s.t. } \mathcal{F} := \{L_1(S) : W(f, \hat{f}) \leq \epsilon\}, \end{aligned}$$

which implies the worst-case distribution f^* (for district P_i 's service) only depends on P_i and $\hat{f}$. By hiding the other terms, we have the inner problem (3) whose solution is what we define as f^* .

$$\begin{aligned} \alpha(P_i) &:= \max_{f \in \mathcal{F}} \beta \sqrt{\Lambda} \int_{P_i} \sqrt{f} \\ & \text{s.t. } \mathcal{F} := \{f \in L_1(S) : W(f, \hat{f}) \leq \epsilon\}. \end{aligned} \quad (3)$$

Expanding ambiguity set $\mathcal{F}$ according to Wasserstein distance, we have the continuous inner problem (4) under the Wasserstein ambiguity.

$$\alpha(P_i) := \max_{f, \pi} \beta \sqrt{\Lambda} \int_{P_i} \sqrt{f} \quad (4a)$$

$$\text{s.t. } \int_S f = 1 \quad (4b)$$

$$\iint_{S \times S} d(\mathbf{x}_1, \mathbf{x}_2) d\pi(\mathbf{x}, \mathbf{y}) \leq \epsilon \quad (4c)$$

$$\int_S \pi(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 = \hat{f}(\mathbf{x}_2), \forall \mathbf{x}_2 \in S \quad (4d)$$

$$\int_S \pi(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_2 = f(\mathbf{x}_1), \forall \mathbf{x}_1 \in S \quad (4e)$$

$$f, \pi \geq 0 \quad (4f)$$

In (4), f denotes the unknown target distribution within the ambiguity set, and π is the probability transport plan that couples f with the nominal distribution $\hat{f}$. The objective (4a) maximizes the aggregated square-root density over district P_i , which captures the in-district travel distance (Corollary 1) under f . Constraint (4b) and (4f) ensure that f is a valid probability density function. Constraint (4c) enforces that the Wasserstein distance between f and $\hat{f}$, measured using metric d , does not exceed the radius ϵ . Finally, constraints (4d) and (4e) guarantee that π has $\hat{f}$ and f as its marginals. Since the Wasserstein ambiguity set $\mathcal{F}$ is compact and convex (Panaretos and Zemel, 2020), the optimal distribution f^* is attainable and (4) is a convex problem with an infinite dimensional decision function f .

Proposition 4 (SDP Formulation for the Inner Worst-Case Distribution Problem (4)). *Given the discrete partition $\{z_{ji}\}$, and the overall arrival rate Λ , $\alpha(P_i)$ in (4) can be approximated by $\hat{\alpha}(\hat{P}_i)$, which is the optimal objective value of the semidefinite program (5).*

$$\hat{\alpha}(\hat{P}_i) := \min_{\rho_j, \lambda, \mu_j, \nu_1, \tau} \beta \sqrt{\Lambda} \left(\sum_{j=1}^N |B_j| \rho_j + \lambda \epsilon + \sum_{j=1}^N \mu_j m_j + \tau \right) \quad (5a)$$

$$s.t. \quad \begin{pmatrix} \rho_j & z_{ji}/2 \\ z_{ji}/2 & \tau + v_j \end{pmatrix} \geq 0, \forall j = 1, \dots, N \quad (5b)$$

$$\lambda d_{jl} + \mu_j - v_l \geq 0, \forall j, l = 1, \dots, N \quad (5c)$$

$$\lambda, \tau, v_j \geq 0, \forall j = 1, \dots, N \quad (5d)$$

In (5), $\{d_{jl}\}_{j,l=1,\dots,N}$ denote the distance between blocks j and l under the chosen metric.

Proof. We first develop a discrete approximation of the inner problem (4), which identifies the discrete worst-case distribution $\tilde{f}$. Then we transform the discrete maximization problem into a dual minimization problem, and show the strong duality holds.

Applying the discretizations, we can reformulate the continuous inner problem (4) into its discrete version (6).

$$\hat{\alpha}(\hat{P}_i) := \max_{x_j, y_{jl}, j, l=1,\dots,N} \beta \sqrt{\Lambda} \sum_{j=1}^N \sqrt{|B_j|} x_j z_{ji} \quad (6a)$$

$$s.t. \quad \sum_{j=1}^N x_j^2 \leq 1 \quad (6b)$$

$$\sum_{j,l=1}^N d_{jl} y_{jl} \leq \epsilon \quad (6c)$$

$$\sum_{l=1}^N y_{jl} = m_j, \forall j = 1, \dots, N \quad (6d)$$

$$\sum_{j=1}^N y_{jl} \geq x_l^2, \forall l = 1, \dots, N \quad (6e)$$

$$y_{jl}, x_j \geq 0, \forall j, l = 1, \dots, N \quad (6f)$$

This discrete formulation (6) corresponds to the continuous inner problem (4), where the adversary selects a density f and transport plan $\pi(\mathbf{x}_1, \mathbf{x}_2)$ over $\mathcal{S} \times \mathcal{S}$ to maximize $\int_{P_i} \sqrt{f(\mathbf{x})} d\mathbf{x}$, subject to normalization (6b), Wasserstein distance (6c), and the marginal consistency (6d), (6e) constraints. In the discrete approximation, f is replaced by a block-uniform histogram with square roots x_j , and π is replaced by the interblock transport plan y_{jl} . The integral $\int_{P_i} \sqrt{f}$ becomes $\sum_j \sqrt{|B_j|} x_j z_{ji}$, the normalization condition becomes $\sum_j x_j^2 \leq 1$, the marginal constraints become $\sum_l y_{jl} = m_j$ and $\sum_j y_{jl} \geq x_l^2$, and the Wasserstein constraint becomes $\sum_{j,l} d_{jl} y_{jl} \leq \epsilon$. This yields a finite problem that approximates the original infinite dimensional formulation (4).

Note that (6) is a convex quadratically constrained problem (CQCP), where the decision variable x_j represents the square root of the probability mass assigned to block B_j by the target distribution f , and y_{jl} (also a decision variable) denotes the transport plan between blocks B_j and B_l .

To obtain an overall minimization formulation, we dualize the primal discrete inner problem (6). For exposition, we omit the constant $\beta \sqrt{\Lambda}$. Then the Lagrangian of (6) is

$$\begin{aligned} \mathcal{L}(x_j, y_{jl}; \tau, \lambda, \mu_j, v_l) &= \sum_{j=1}^N \sqrt{|B_j|} x_j z_{ji} + \tau \left(1 - \sum_{j=1}^N x_j^2 \right) + \lambda \left(\epsilon - \sum_{j,l=1}^N d_{jl} y_{jl} \right) + \sum_{j=1}^N \mu_j \left(m_j - \sum_{l=1}^N y_{jl} \right) + \sum_{l=1}^N v_l \left(\sum_{j=1}^N y_{jl} - x_l^2 \right), \\ &= \sum_{j=1}^N \left(-(\tau + v_j) x_j^2 + \sqrt{|B_j|} z_{ji} x_j \right) + \sum_{j,l=1}^N (v_l - \mu_j - \lambda d_{jl}) y_{jl} + \lambda \epsilon + \sum_{j=1}^N \mu_j m_j + \tau. \end{aligned}$$

Since $\sum_j x_j^2 \leq 1$, $\sum_{j,l} d_{jl} y_{jl} \leq \epsilon$, and $\sum_j y_{jl} \geq x_l^2$, the dual variables $\tau, \lambda, v_l \geq 0, \forall l = 1, \dots, N$. Then consider maximizing $\mathcal{L}$ over $x_j, y_{jl} \geq 0$. If there exists a pair (j, l) such that $v_l - \mu_j - \lambda d_{jl} \geq 0$, the maximization is

unbounded. Hence, the dual is constrained by $\lambda d_{jl} + \mu_j - \nu_l \geq 0, \forall j, l = 1, \dots, N$ and at the optimal solution, $y_{jl}^* = 0, \forall j, l = 1, \dots, N$. Then we have

$$\max_{x_j \geq 0} \mathcal{L}(x_j, y_{jl}^*; \tau, \lambda, \mu_j, \nu_l) = \sum_{j=1}^N \left(-(\tau + \nu_j) x_j^2 + \sqrt{|B_j| z_{ji}} x_j \right) + \lambda \epsilon + \sum_{j=1}^N \mu_j m_j + \tau.$$

Notice when $\tau + \nu_j = 0$, $\max_{x_j \geq 0} \mathcal{L}$ is unbounded. As a result, the dual is also constrained by $\tau + \nu_j > 0, \forall j : z_{ji} = 1$. Hence, $\mathcal{L}$ is quadratically concave in each x_j with $z_{ji} = 1$ for district P_i , and each $x_j^* = \sqrt{|B_j| z_{ji}} / (2(\tau + \nu_j))$. Plugging x_j^* back yields the dual objective

$$\min_{\tau, \lambda, \nu_l \geq 0} \mathcal{L}(x_j^*, y_{jl}^*; \tau, \lambda, \mu_j, \nu_l) = \min_{\tau, \lambda, \nu_l \geq 0} \sum_{j=1}^N \frac{|B_j| z_{ji}^2}{4(\tau + \nu_j)} + \lambda \epsilon + \sum_{j=1}^N \mu_j m_j + \tau.$$

Given $\tau + \nu_j > 0$, by Schur complement we have $\rho_j \geq z_{ji}^2 / (4(\tau + \nu_j))$ is equivalent to the LMI

$$\begin{pmatrix} \rho_j & z_{ji}/2 \\ z_{ji}/2 & \tau + \nu_j \end{pmatrix} \geq 0.$$

Hence, we can replace $z_{ji}^2 / (4(\tau + \nu_j))$ with ρ_j . Also notice that for any $j : z_{ji} = 1$, the minimization of $\mathcal{L}(x_j^*, y_{jl}^*; \tau, \lambda, \mu_j, \nu_l)$ over $(\tau \geq 0, \lambda \geq 0, \mu_j, \nu_l \geq 0)$ automatically implies $\tau^* + \nu_j^* > 0$ because $|B_j| z_{ji}^2 > 0$. And for $j : z_{ji} = 0$, $\tau + \nu_j$ is free. Hence, we can also relax the constraints $\tau + \nu_j > 0, \forall j : z_{ji} = 1$ into $\tau + \nu_j \geq 0, \forall j = 1, \dots, N$, which is already implied by $\tau, \nu_j \geq 0, \forall j = 1, \dots, N$. Finally, the dualized problem is (5) after multiplying the constant $\beta \sqrt{\Lambda}$ back, and by inspection it is an semidefinite program (SDP).

To show strong duality, we can find a Slater point in the primal feasible set. Notice that only constraints $\sum_j x_j^2 \leq 1$ and $\sum_j y_{jl} \geq x_l^2, \forall l$ are not affine, so we only need to find $x_j, y_{jl}, \forall j, l$ such that inequalities are strict. One such construction exists: $x_j = 0, y_{jl} = \epsilon / (N^2 d_{jl})$ if $j \neq l$ otherwise 0. Therefore, strong duality holds. $\square$

Moreover, We can quantify the error induced by two discretizations: replacing the true region P_i with its block-union approximation $\hat{P}_i$, and replacing the target f with its block-wise histogram $\tilde{f}$. In particular, we will bound $|\alpha(P_i) - \hat{\alpha}(\hat{P}_i)|$. Theorem (5) makes this precise.

Proposition 5 (Discretization Error of $\hat{\alpha}(\hat{P}_i)$). *Let δ be the maximum diameter of all blocks, i.e. $\delta := \max_j \text{diam}(B_j)$. For each district $P_i \subset S$, define its corresponding block-union approximation $\hat{P}_i = \cup_{j: B_j \subset P_i} B_j$. Assume probability density f is Lipschitz on S and let $\alpha(P_i) = \beta \sqrt{\Lambda} \int_{P_i} \sqrt{f(\mathbf{x})} d\mathbf{x}$, $\hat{\alpha}(\hat{P}_i) = \beta \sqrt{\Lambda} \int_{\hat{P}_i} \sqrt{\tilde{f}(\mathbf{x})} d\mathbf{x}$, where $\tilde{f}$ is the blockwise uniform histogram matching each block's mass of f . Then $|\alpha(P_i) - \hat{\alpha}(\hat{P}_i)| = O\left(M_{\sqrt{f}} |\partial P_i| \delta + L_{\sqrt{f}} |S| \delta\right) = O(\delta)$, i.e. the approximation error on $\alpha(P_i)$ is $O(\delta)$.*

The proof of Proposition 5 is in Appendix D. In essence, we split the total error into the partition error the distribution error, and we show each piece is $O(\delta)$. Hence, summing these two bounds yields an overall discretization error of order $O(\delta)$, which goes to zero as the maximum block diameter δ vanishes. This formalizes the intuition that, by refining the atomic blocks, the discrete approximation $\hat{\alpha}(\hat{P}_i)$ converges to the true worst-case objective $\alpha(P_i)$. This coincides with the intuition in Steele (1981), where the piecewise uniform case is extended to the nonuniform case using similar approximations.

4.3. Discrete Overall Model

To incorporate the dual discrete inner problem (5) into the overall decision framework, we adopt a unified indexing scheme for districts and blocks. In this scheme, the index of each potential district is aligned with the index of its designated root block. Consequently, the same index set $\{1, \dots, N\}$ is used to label both the blocks $\{B_j\}$ and the districts $\{P_i\}$, and a district P_i is considered active only when its root block B_i is selected, that is, when $z_{ii} = 1$.

Table 1
Summary of Notations.

Parameters	
$\{B_j\}$	Set of blocks / elements of S after discretization
k	Number of districts
N	Number of blocks
m_j	Nominal probability mass at block B_j
κ_j	Centroid of block B_j
$\phi(\kappa_j)$	Depot setup cost at location κ_j
l	Per-AV fixed cost
Q	AV capacity
s	Speed of AVs
ϵ	Wasserstein radius of ambiguity
d^{ODD}	Dimensionality of ODD feature vectors
Ω_j	ODD feature vector of block B_j
η_j	Cost weight for ODD feature j
Λ	Total demand arrival rate
d_{ij}	Distance between B_i and B_j under chosen metric
$\beta \approx 0.7120$	Constant in BHH TSP approximation
w_r	Cost weight of user cost
ϵ	Probability of controlled events in chance constraints
$\bar{\lambda}(Q, \epsilon)$	Maximum arrival rate such that $\mathbb{P}(\text{number of demands} \geq Q) \leq \epsilon$
$\mathcal{G}$	Graph of blocks, arcs constructed by adjacency
$ B_j $	Area of block B_j
$A(\mathcal{G})$	Set of all arcs on graph $\mathcal{G}$
Decision Variables	
δ_j	Indicator of whether depot is at B_j
z_{ji}	Indicator of whether block B_j is assigned to district with root B_i
c	Auxiliary decision for the maximum district-level cost
K_i	Linehaul cost of district with root B_i
F_i	ODD cost of district with root B_i
V_i	Fleet size assigned to district with root B_i
T_i	Dispatch interval in district with root B_i
x_{ji}	The sqrt mass of worst-case distribution for district with root B_i
$\rho_{ji}, \lambda_i, \mu_{ji}, \nu_{li}, \tau_i$	Dual decisions for inner problem
O	Overall ODD cost
ω_i	ODD summary feature vector of district with root B_i
f_{ij}^v	Flows sent from B_v on arc (i, j)
$\bar{Q}_i$	Equivalent AV capacity ensuring returning before next dispatch on district with root B_i

This convention provides a consistent representation of district-level decisions and establishes a direct correspondence between each district and its root block location.

With this unified indexing, block B_i serves as the root of district P_i , and the assignment variables z_{ji} indicate whether block B_j belongs to district P_i . To ensure that each district forms a single contiguous region, we impose flow-based contiguity constraints following Validi et al. (2022). For each district, a unit of flow originates at its root block B_i and must reach every block assigned to P_i , using only arcs that traverse blocks with $z_{ji} = 1$. These flow

constraints guarantee that all assigned blocks are connected through admissible paths, thereby enforcing contiguity in the constructed districts.

Because problem (5) computes the worst-case cost for one district at a time, the full model contains a separate copy of all its decision variables $(\rho_{ji}, \lambda_i, \mu_{ji}, v_{ji}, \tau_i)$ for each district P_i . Finally, by coupling these indexed inner problem decisions with the depot location $\{\delta_j\}$, partition $\{z_{ji}\}$, fleet sizing $\{V_i\}$, dispatch intervals $\{T_i\}$, and other auxiliary decisions denoting costs, we arrive at the complete mixed-integer nonlinear problem in (7), where κ_j and Ω_j are the location centroid and the maximum ODD summary vector of block B_j respectively.

$$\min_{\substack{\delta_j, z_{ji}, c, K_i, F_i, V_i, T_i, \\ x_{ji}, \rho_{ji}, \lambda_i, \mu_{ji}, v_{ji}, \tau_i, \\ \bar{Q}_i, O, \omega_i, f_{ij}^v}} \sum_{j=1}^N \phi(\kappa_j) \delta_j + l \sum_{i=1}^N V_i + O + c \quad (7a)$$

$$\text{s.t. } \delta_j, z_{ji} \text{ binary, } V_i \text{ nonnegative integer, } \forall i, j = 1, \dots, N \quad (7b)$$

$$\sum_{j=1}^N \delta_j = 1 \quad (7c)$$

$$\sum_{i=1}^N z_{ji} = 1, \forall j = 1, \dots, N \quad (7d)$$

$$\sum_{i=1}^N z_{ii} = k \quad (7e)$$

$$z_{ji} \leq z_{ii}, \forall i, j = 1, \dots, N \quad (7f)$$

$$K_i = 2 \sum_{j=1}^N d_{ij} \delta_j, \forall i = 1, \dots, N \quad (7g)$$

$$\omega_i \geq \Omega_j z_{ji}, \forall j = 1, \dots, N \quad (7h)$$

$$F_i = \sum_{j=1}^{d^{\text{odd}}} \eta_j \cdot (\omega_i)_j, \forall i = 1, \dots, N \quad (7i)$$

$$O \geq F_i, \forall i = 1, \dots, N \quad (7j)$$

$$x_{ji} = \frac{\sqrt{|B_j|} z_{ji}}{2(\tau_i + v_{ji})}, \forall i, j = 1, \dots, N \quad (7k)$$

$$V_i \geq \frac{\Lambda T_i}{\bar{\lambda}(Q, \epsilon)} \sum_{j=1}^N x_{ji}^2, \forall i = 1, \dots, N \quad (7l)$$

$$\bar{Q}_i = \frac{(sT_i - K_i)^2}{\beta^2} \frac{\sum_{j=1}^N x_{ji}^2}{\left(\sum_{j=1}^N x_{ji}\right)^2}, \forall i = 1, \dots, N \quad (7m)$$

$$V_i \geq \frac{\Lambda T_i}{\bar{\lambda}(\bar{Q}_i, \epsilon)} \sum_{j=1}^N x_{ji}^2, \forall i = 1, \dots, N \quad (7n)$$

$$f^i(\delta^-(j)) - f^i(\delta^+(j)) = z_{ji}, \forall i, j = 1, \dots, N, i \neq j \quad (7o)$$

$$f^i(\delta^-(j)) \leq (N-1)z_{ji}, \forall i, j = 1, \dots, N, i \neq j \quad (7p)$$

$$f^i(\delta^-(i)) = 0, \forall i = 1, \dots, N \quad (7q)$$

$$f_{ij}^v \geq 0, \forall (i, j) \in A(\mathcal{G}), \forall v = 1, \dots, N \quad (7r)$$

$$c \geq \beta \sqrt{\Lambda} \left(\frac{1}{\sqrt{T_i/V_i}} + \frac{w_r}{s} \sqrt{T_i/V_i} \right) \left(\sum_{j=1}^N |B_j| \rho_{ji} + \lambda_i \epsilon + \sum_{j=1}^N \mu_{ji} m_j + \tau_i \right) + z_{ii} \left(\frac{K_i}{T_i/V_i} + \frac{w_r K_i}{2s} + w_r T_i/V_i \right), \forall i = 1, \dots, N \quad (7s)$$

$$\begin{pmatrix} \rho_{ji} & z_{ji}/2 \\ z_{ji}/2 & \tau_i + v_{ji} \end{pmatrix} \geq 0, \forall i, j = 1, \dots, N \quad (7t)$$

$$\lambda_i d_{ji} + \mu_{ji} - v_{li} \geq 0, \forall i, j, l = 1, \dots, N \quad (7u)$$

$$\lambda_i \geq 0, \tau_i \geq 0, v_{ji} \geq 0, \forall i, j = 1, \dots, N \quad (7v)$$

The full mixed-integer nonlinear problem (MINLP) (7) jointly determines the depot location, the regional partition, the dispatch intervals, and the fleet sizes under the worst-case demand distributions. The depot location is encoded by the binary variables $\delta_{1:N}$, where $\delta_j = 1$ indicates that the depot is placed in block B_j . The regional partition is represented by the assignment variables $z_{1:N,1:N}$: each block B_j is assigned to exactly one district rooted at block B_i when $z_{ji} = 1$, and a district P_i is activated when its root block satisfies $z_{ii} = 1$. Through these assignments, $z_{1:N,1:N}$ also determines the linehaul distance K_i and the ODD cost F_i for each district, since both depend on the spatial configuration of the blocks assigned to P_i . The system-level ODD cost O is defined as the maximum district ODD cost across all active districts. The constraints governing these decisions are specified in (7b), (7c), (7d), (7e), (7f), (7g), (7h), (7i), and (7j).

To ensure that no AV is overloaded beyond its capacity Q and that every AV returns to the depot before its next dispatch with probability at least $1 - \epsilon$, we translate the continuous chance constraints (2k), (2m), and (2l) into their discrete analogues (7k), (7l), (7m), and (7n). The derivation relies on the block-based representation of the worst-case distribution obtained from the dual of the inner problem. In particular, the variable x_{ji} represents the square root of the probability mass assigned to block B_j inside P_i by the worst-case distribution, and takes the form

$$x_{ji} = \frac{\sqrt{|B_j|} z_{ji}}{2(\tau_i + v_{ji})}, \forall i, j = 1, \dots, N. \quad (10k)$$

This expression follows from the KKT stationary conditions on the primal (6) and dual (5), where τ_i and v_{ji} are the dual multipliers associated with the normalization and marginal-consistency constraints of the Wasserstein ambiguity set.

Given this representation, the total worst-case mass assigned to district P_i becomes $\sum_{j=1}^N x_{ji}^2$, which discretizes the integral $\int_{P_i} f$. Substituting this discrete mass into the continuous fleet-size lower bound derived from the capacity chance constraint yields (7l), which is the discrete counterpart of (2k). Similarly, the on-time return requirement in (2m) involves the quantity $\bar{Q}_i$, the largest number of users that can be served within a dispatch while still returning to the depot on time. Discretizing its continuous expression leads to (7m), which captures both the fixed linehaul distance K_i and the in-district TSP travel term through the block-based density representation. Substituting $\bar{Q}_i$ into the fleet-size lower bound associated with the return-time chance constraint produces (7n) which corresponds to (2m). Together, these discrete formulations preserve the probabilistic guarantees of the continuous chance constraints while embedding the worst-case distribution directly into the block-based MINLP through the dual variables.

To guarantee each district forms a single contiguous region so that the vehicle can always travel from one location to another, we embed the flow-based contiguity constraints from Validi et al. (2022). Specifically, a directed planar graph $\mathcal{G}$ with the blocks as vertices is constructed and if block B_i is adjacent to block B_j , $(i, j) \in A(\mathcal{G})$, where $A(\mathcal{G})$ is the arc set of $\mathcal{G}$. Let $\delta^+(i)$ be the set of arcs from block B_i , and $\delta^-(i)$ be the set of arcs to block B_i . The contiguity constraints are then imposed by flows from the root block B_i of each district to all the blocks in the same district through the arcs, and each non-root block B_j consumes one unit of such flow if $z_{ji} = 1$. For notation ease, f_{ij}^v denotes the amount of flow originating from district root B_v across the arc (i, j) , and $f^j(S) := \sum_{(u,v) \in S} f_{uv}^j$. The resulting flow-conservation and capacity constraints (7o), (7p), (7q), (7r) prevent disconnected clusters, ensuring that each district is a contiguous area suitable for deployment.

The auxiliary decision c denotes the maximum district-level cost (excluding the depot cost) by constraint (7s). The inner discrete problem (6) is replaced by its SDP dual (5) with decisions $(\rho_{ik}, \lambda_i, \mu_{il}, v_{ik}, \tau_i)$, producing a tractable set of 2×2 LMIs ((7t)) and linear inequalities ((7u), (7v)).

Since each district's cost objective is linear in $\hat{\alpha}$ in (7s), hence a vanishing error on $\hat{\alpha}$ also leads to the vanishing error on the district-level costs, given that the objective of the inner problem (3) is the only part depending on the partition and distribution among all terms in the overall objective (2a).

4.4. Search Heuristic

We design a lightweight heuristic inspired by Resende and Werneck (2004) and Validi et al. (2022) that combines randomized exploration with a greedy local search. A central building block of the heuristic is the modified compactness model (HessModel), based on the classical framework of Hess, Weaver, Siegfeldt, Whelan and Zitlau (1965). Unlike the original Hess model, which determines district centers as part of the optimization, our version takes a prescribed set of k district roots S as input. We enforce this through the constraints $z_{ii} = 1$ for $i \in S$ and $z_{ii} = 0$ for $i \notin S$, ensuring that assignments are made only to the selected roots. Solving (HessModel) for a given S then yields a compact and coherent district assignment that respects the chosen roots, making it a fast and reliable subroutine for both stages of the heuristic.

$$\begin{aligned} \text{HessModel}(\mathcal{G}, S) := \min_{z_{ji}} \quad & \sum_{i,j=1}^N d_{ji} z_{ji} \\ \text{s.t.} \quad & \sum_{i=1}^N z_{ji} = 1, \forall j = 1, \dots, N \\ & z_{ji} \leq z_{ii}, \forall i, j = 1, \dots, N \\ & z_{ii} = 1, \forall i \in S \\ & z_{ii} = 0, \forall i \notin S \\ & z_{ji} \text{ binary}, \forall i, j = 1, \dots, N \end{aligned} \quad (\text{HessModel})$$

Given any depot location D , district root set S , and block assignments $\mathbf{z}$, we define $\Phi(D, S, \mathbf{z})$ as the overall cost objective value with the optimized fleet sizes, dispatch intervals under the induced worst-case demand distributions. This enables us to evaluate the quality of any candidate configuration. The heuristic operates in two stages. In the random exploration phase, the algorithm repeatedly samples a depot location and a root set S , constructs the corresponding district assignments by solving (HessModel), and relocates the roots to reduce assignment distances. This phase provides broad diversification by generating partitions that explore different regions of the solution space. The second stage is a recursive local search, which focuses on refining a promising configuration. For each root $s \in S$, the algorithm considers replacing it with a neighboring block s' in the block graph, then solves (HessModel) to update the assignments and evaluates the resulting configuration using Φ . If an improving move is found, the procedure recurses on the updated configuration. The search proceeds until no improving root relocation exists. When combined with the random exploration phase, this local refinement produces a robust balance between global search and localized improvement. Algorithms 1 and 2 present the full procedure. The evaluation of $\Phi(D, S, \mathbf{z})$ is in Appendix C.

5. Case Study

5.1. Setting: Past and Present of Eastern Pittsburgh

We apply our service design in the context of a nonprofit last-mile transportation service provider referred as X, in the Greater Pittsburgh region. They provide essential service for the East Pittsburgh, Monroeville, and McKeesport communities, east of Pittsburgh city, where residents have low household income, low vehicle ownership rate, and lack direct access to public transit.

These eastern Pittsburgh communities were shaped by the region's industrial past, when steel mills, rail lines, and river-adjacent industrial parks anchored local jobs and transit. Neighborhoods grew tightly along hillsides above the mills, producing the narrow, steep, and winding streets that still define mobility challenges today. When industry declined, these areas lost jobs, tax bases, and much of the transit service that once followed rail and mill worker routes. The result is a uniquely constrained operational design domain: steep topography, limited street widths, rails, fragmented neighborhoods, and irregular transitions between industrial and residential areas. These physical conditions make traditional high-frequency transit difficult to operate and maintain.

Algorithm 1 Local Search

Require: Block graph $\mathcal{G}$, depot block D , district roots S **Ensure:** Improved depot location D , district roots S , and assignments $\mathbf{z}$

```
1:  $\mathbf{z} \leftarrow \text{HessModel}(\mathcal{G}, S)$ 
2: for each root  $s \in S$  do
3:   for each neighbor  $s' \in N_{\mathcal{G}}(s) \setminus S$  do
4:      $S' \leftarrow (S \setminus \{s\}) \cup \{s'\}$ 
5:      $\mathbf{z}' \leftarrow \text{HessModel}(\mathcal{G}, S')$  restricted to roots in  $S'$ 
6:      $D' \leftarrow$  the depot location minimizing the average distance to all districts
7:     if  $\Phi(D', S', \mathbf{z}') < \Phi(D, S, \mathbf{z})$  then
8:       return LocalSearch( $\mathcal{G}, D', S'$ )
9:     end if
10:  end for
11: end for
12: return  $D, S, \mathbf{z}$ 
```

Algorithm 2 Combined Random Search and Local Search

Require: Block graph $\mathcal{G}$, number of districts k , iteration limit max_iterations**Ensure:** Final depot location D^* , set of k district roots S^* and assignments $\mathbf{z}^*$

```
1: Initialize best objective  $\Phi^* \leftarrow +\infty$ , best depot location  $D^* \leftarrow \text{None}$ , best roots  $S^* \leftarrow \emptyset$ 
2: for  $t = 1, \dots, \text{max\_iterations}$  do
3:   Sample one depot location  $D \in V(\mathcal{G})$  and root set  $S \subseteq V(\mathcal{G})$  with  $|S| = k$  uniformly at random
4:   repeat
5:     Compute assignment  $\mathbf{z} \leftarrow \text{HessModel}(\mathcal{G}, S)$ 
6:      $S' \leftarrow$  relocate the roots by minimizing the distance from assigned blocks to their roots
7:      $S \leftarrow S'$ 
8:   until convergence
9:    $D \leftarrow$  the depot location minimizing the average distance to all districts
10:   $\Phi \leftarrow \Phi(D, S, \mathbf{z})$ 
11:  if  $\Phi < \Phi^*$  then
12:     $\Phi^* \leftarrow \Phi$ ,  $D^* \leftarrow D$ ,  $S^* \leftarrow S$ 
13:  end if
14: end for
15:  $(D^*, S^*, \mathbf{z}^*) \leftarrow \text{LocalSearch}(\mathcal{G}, D^*, S^*)$ 
16: return  $D^*, S^*, \mathbf{z}^*$ 
```

A persistent driver shortage, intensified by competition from commercial delivery services and the demands of navigating complex terrain, further strains X's operational capacity. In addition, surveys from X's communities persistently cite location mismatch as a reason for their low usage of X fixed-route shuttles, despite its low fare (25 cents per ride). At the time of our collaboration with X, we saw increasing operational cost and decreasing ridership after COVID, resulting in the doubling of their "cost per rider" performance metric. This has led to increased but perhaps unwarranted scrutiny towards their operational excellence by their board of directors, threatening reduced funding and a vicious cycle of even less operational capacity and lower ridership. This combination of industrial legacy, difficult geography, staffing constraints, operational challenges, and demand mismatch highlights the need for innovative last-mile service design. We were tasked to think creatively to design future service modes that take into considerations of the operational design challenges, demand variations, and driver shortage.

Figure 5 shows X's current operations: three fixed routes with one shuttle each (left), and the corresponding partition (right): McKeesport district in red, Monroeville in yellow, and East Pittsburgh in gray. All three originate at a Walmart hub and follow predetermined loops that cover their assigned districts before returning to the hub. The routes are contiguous and do not cross district boundaries. This partitioned structure simplifies daily operations and dispatch

planning. However, it also restricts flexibility and makes the system highly sensitive to spatial mismatches between the fixed routes and the travel demand locations.

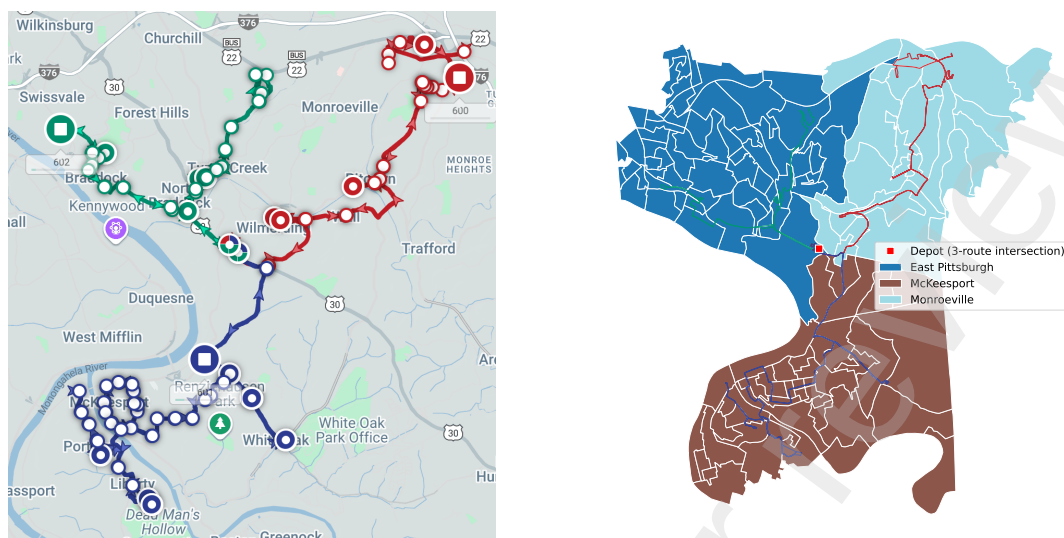


Figure 5: Partitioned Operations of X.

X has three routes, each responsible for a particular district (East Pittsburgh, Monroeville, McKeesport).

Figure 6 shows that rider demand varies considerably across both time and space. Areas with the highest demand in the morning are not the same as those with the highest demand in the afternoon, and weekday demand patterns differ markedly from those observed on Saturdays. Because X cannot reconfigure districts and redistribute AVs from hour to hour for each new demand pattern, relying on scenario-dependent recourse is impractical. This pronounced variability underscores the need for a service plan that is robust to a wide range of demand patterns. By optimizing against the worst-case distribution within an ambiguity set, our DRO framework is well suited to produce designs that remain effective even under substantial spatial-temporal fluctuations in demand.

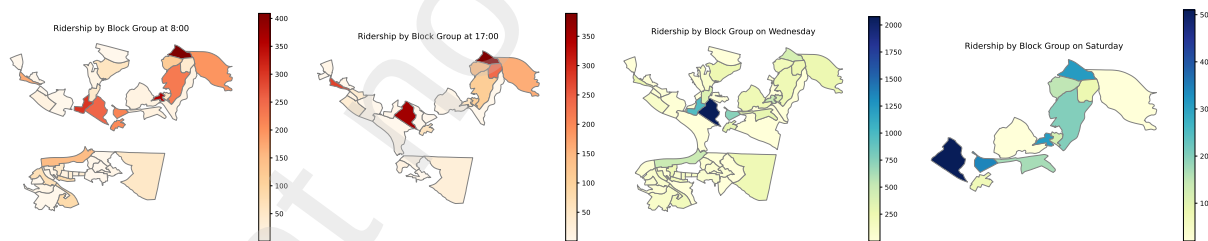


Figure 6: Demand Distribution at Census Block Group Level

Demand distributions at 8:00am and 17:00, on Wednesdays and Saturdays.

In addition, X's existing fixed-route system exhibits a workload imbalance across districts. As shown in Figure 7, Monroeville consistently serves nearly half of all annual trips, while other districts carry substantially lighter loads. This structural asymmetry leads riders in high-load districts to face longer wait times and more variable service quality. To address these inequities, our objective adopts a min-max criterion that minimizes the worst district-level cost.

5.2. Envisioning A Future With Demand-Responsive Autonomous Transit

Building on X's interest in reimagining a flexible and technology-enabled service design to enable affordable and responsive service, our framework introduces the demand-responsive last-mile service supported by autonomous vehicles. In this service mode, users register their destinations at the central hub or through pre-booking at their arrivals. This information allows the AVs to plan each dispatch route with more detailed and timely knowledge of

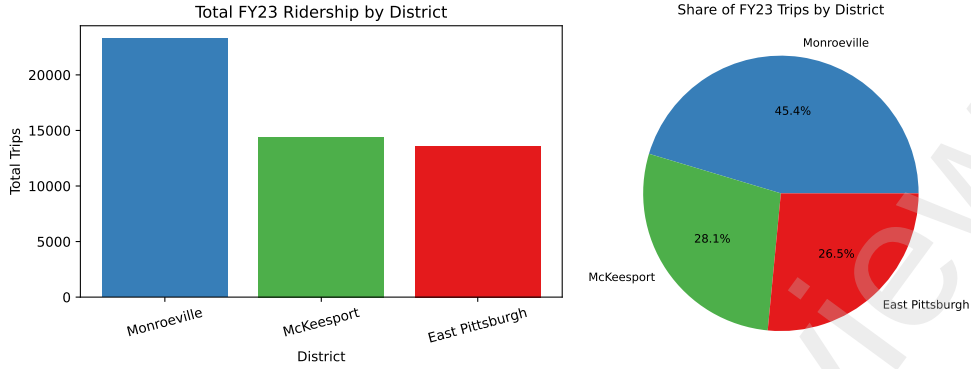


Figure 7: Imbalanced Ridership in 2023 Fiscal Year

demand. Rather than following fixed routes, the vehicles depart from the depot and transport users to their requested destinations, saving traveling time and costs.

We discretize the service region into census block groups (BGs) (U.S. Census Bureau, 2023), which serve as the fundamental geometric units for districting and demand allocation. Each block group is assumed to have uniform demand density, and inter-block distances are computed using Euclidean distances between BG centroids. This approximation is appropriate for East Pittsburgh, where the roadway network differs from a Manhattan-style grid and the Euclidean metric captures spatial relationships reliably at the BG scale. To construct the ambiguity set for worst-case demand, we start from X's Spring 2023 ridership dataset, which contains detailed timestamped destination information at the block-group level.

ODD features are incorporated following the ISO technical specification for automated driving system operating conditions (International Organization for Standardization, 2025). The ISO framework covers a broad range of roadway, infrastructure, environmental, and interaction-related attributes. From this list, we select the subset of features for low-speed last-mile autonomous vehicle service and that can be reliably computed using publicly available data, and construct an ODD feature dataset. Table 2 summarizes the full ISO feature taxonomy and indicates which features are included in our implementation.

Our selected ODD features fall into three categories: road geometry (curvature, grade, number of lanes), static infrastructure (railway adjacency, tunnel presence), and environmental context (elevation). These features were chosen because they (i) directly affect CAV maneuverability and routing feasibility, (ii) vary meaningfully across X's roadway network, and (iii) can be consistently computed at the census block group level.

To operationalize these ODD features, we compute each block group's feature vector through a sequence of geospatial and network-processing steps. For each block group B_j , we first extract its polygon geometry from the 2023 Census TIGER/Line block-group shapefile (U.S. Census Bureau, 2023) and compute the mean elevation by overlaying the polygon on the 1-arcsecond SRTM digital elevation model (DEM) (NASA JPL, 2013) and averaging all raster values within the polygon. We then download the full road network contained inside the union of $\{B_j\}$ composing the whole service region S from OpenStreetMap (OSM) (OpenStreetMap contributors, 2017) using a polygon-based query and convert each road segment into a metrically projected linestring. Road curvature is computed as the sum of turning angles along each linestring divided by its length, and road grade is obtained by sampling the DEM at the endpoints of each segment and computing the rise-over-run ratio in meters. Additional roadway attributes such as number of lanes, posted or inferred speed limit, road classification, and bridge and tunnel indicators are extracted directly from OSM tags. Rail presence is computed by clipping the TIGER/Line rail shapefile to the study-area bounding box and then, for each block-group polygon, testing whether any of those clipped rail geometries intersect it.

After computing these features at the road-segment level, we perform a spatial join that assigns each segment to a block group B_j and aggregate the resulting values. Curvature, grade, and speed are averaged within each B_j , binary attributes such as bridge, tunnel and rail presence are summarized using logical maxima. Road-type difficulty is obtained by mapping OSM highway tags to ordinal difficulty levels (ranked in Table 4) and selecting the most challenging type present in B_j . This procedure yields a numerical feature vector ϕ_{B_j} that characterizes the roadway geometry, static infrastructure, and environmental context of block group B_j .

These block-group-level feature vectors form the rows of the matrix Ω , which serves as the input to our ODD cost formulation. For any district P_i , the associated ODD vector ω_j is constructed by applying the elementwise maximum operator across the set $\{\omega_{B_j} : B_j \subseteq P_i\}$ (i.e. $\{\omega_{B_j} : z_{ji} = 1\}$) so that the district inherits the most challenging environmental conditions present in its constituent block groups. The resulting summary vector ω_i enters the modular cost function $\mathcal{J}(\omega_i)$, closing the connection between the geospatial computations described in this section and the ODD-based operational cost used in the optimization model.

Figure 8 illustrates an example how ODD features are extracted from the road environment. The left panel shows some part of the road network, and the highlighted segment corresponds to the zoomed-in satellite image on the right. This specific segment contains a bridge crossing Turtle Creek, exhibits some curvature, and lies adjacent to an active railway corridor. These physical and contextual characteristics translate into the ODD feature vector displayed in the figure, demonstrating how raw geospatial data are mapped into numerical descriptors used by our model.

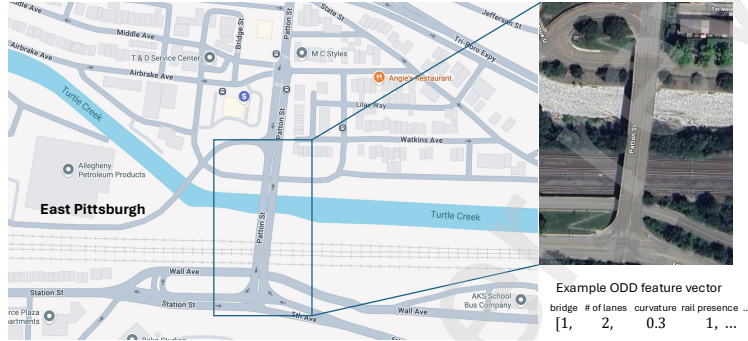


Figure 8: ODD Feature Mapping Example

Because these ODD features serve as exogenous environmental factors that influence road difficulty rather than demand uncertainty, they enter our model as block-group-level parameters that inform the partitioning step. The constructed ODD feature dataset provides a granular, interpretable representation of the physical and infrastructure environment in which AVs operate.

Table 3 presents the estimated cost coefficients for the nine selected ODD features. These coefficients reflect the marginal challenge each feature introduces to autonomous vehicle operations. Features such as curvature, grade, and speed limit capture roadway geometry and motion dynamics, while tunnel presence and bridge crossings indicate environments with potential occlusions and perception challenges. For road classification, we assign coefficients by type, increasing with complexity and decreasing AV predictability. As shown in Table 4, road types are ordered from busways and motorways with minimal uncertainty to unclassified and dense residential streets where AVs face the greatest navigation difficulty. These estimates synthesize domain heuristics and existing engineering assessments to support robust AV deployment planning.

5.3. Result: Reassigning A Few Blocks Leads to Significant Cost Saving

Using this setup, we solve the demand-responsive design via Formulation (7) and Algorithm 2, computing a distributionally robust design by minimizing worst-case costs over a Wasserstein set centered at the commuting demand distribution derived from 2023 Spring data.

We compare three service designs: (i) $\mathcal{P}_0$ FR, a fixed-route service under the nominal partition; (ii) $\mathcal{P}_0$ DR, a demand-responsive service under the nominal partition; and (iii) $\mathcal{P}^*$ DR, a demand-responsive service under the robustly optimized partition. Table 5 reports two classes of performance metrics. User metrics quantify walking, waiting, and in-vehicle transit times and are expressed in minutes. Provider metrics reflect the operator's routing workload and are expressed as total travel distance per hour (km/h). This km/h value is not a vehicle speed; instead, it represents the amortized vehicle-kilometers that must be supplied each hour in steady-state service.

For user metrics we report: Expected (under the nominal distribution), Worst-IP (expected performance under the worst-case distribution), and Worst-user (the maximum individual total cost). In the fixed-route design, the Worst-user column includes the maximum individual walk time because users need to walk to the nearest stop. In demand-responsive designs (which remove walking), the same column reflects the maximum individual wait-plus-transit time

Table 2

Operational Design Domain (ODD) Features for Road Segments (International Organization for Standardization, 2025)

Feature	Example Value	Feature Type	Used?
Road Geometry			
Curvature	0.015 rad	Continuous	✓
Grade / slope	2%	Continuous	✓
Lane width	3.2 meters	Continuous	X
Number of lanes	2	Discrete–Ordinal	✓
Presence of median or divider	None	Binary	X
Road Type and Classification			
Road classification	Arterial road	Discrete–Categorical	✓
Speed limit	35 mph	Discrete–Ordinal	✓
Controlled access	No	Binary	X
Intersection type	Merge / signalized intersection	Discrete–Categorical	X
Bridge presence	Bridge crossing Turtle Creek	Binary	✓
Static Infrastructure Features			
Railway presence	Yes	Binary	✓
Tunnel presence	No	Binary	✓
Shoulder width	0.8 meters	Continuous	X
Road surface quality	Medium	Discrete–Ordinal	X
Environmental Context			
Elevation	282 meters	Continuous	✓
Lighting level	Medium	Discrete–Ordinal	X
Vegetation / occlusion level	High	Discrete–Ordinal	X
Surrounding land use	Mixed industrial / residential	Discrete–Categorical	X
Traffic and Interaction Complexity			
Historical vehicle density	Medium	Discrete–Ordinal	X
Pedestrian presence likelihood	Low	Discrete–Ordinal	X
Cyclist presence likelihood	Low	Discrete–Ordinal	X

Table 3

Estimated ODD Feature Cost Coefficients

ODD Feature	Coefficient (η_j)	Notes
Curvature	5.0	Safety risk increases with curvature
Grade / slope	3.0	Affects energy use and AV braking performance
Number of lanes	2.0	More lanes cause more interaction complexity
Road classification	Table 4	Arterials and collectors differ in AV readiness
Speed limit	2.5	Higher speeds require higher perception + reaction
Bridge presence	4.5	Special handling, occlusions, structure risk
Railway presence	2.0	Potential AV confusion near active rails
Tunnel presence	6.0	Sensor occlusion, GPS loss
Elevation	1.0	Minimal impact on AVs; used as a proxy

under the worst-case distributions across districts. The provider metrics include Worst and Expected amortized travel distances per hour, computed analogously under the worst-case and nominal demand distributions.

Table 5 summarizes the performance of the three service modes and decomposes their user and provider costs. The fixed-route design $\mathcal{P}_0$ FR exhibits high user time because they may need to walk to predetermined stops and experience long in-vehicle travel due to circuitous loops. Demand-responsive service under the same partition $\mathcal{P}_0$ DR eliminates walking and substantially reduces both expected and worst-case user costs. Provider costs also decline because 1) AVs travel only to requested destinations rather than following fixed loops, 2) AV's per-mile cost is lower compared to human drivers.

Table 4
Road Type and Cost Coefficients

Road Type	Coefficient (η_j)	Notes
Busway	1.0	Controlled lanes, minimal cross traffic or surprises
Motorway	1.5	High-speed but fully grade separated, consistent geometry
Trunk	3.0	Expressways that can mix access/traffic types, a bit more complex than motorways
Primary	5.5	Major arterials with signals, pedestrians, and turning traffic
Secondary	9.0	Mid-level arterials; more local access and variability
Tertiary	13.5	Collectors feeding neighborhoods; frequent driveways and mixed users
Residential	19.0	Dense local streets, parked cars, pedestrians/cyclists everywhere
Unclassified	25.0	Catch-all for alleys or service roads; least predictable, sparse mapping detail

Table 5
Comparison of Travel Time and Cost

Design		User Cost (mins)			Provider Cost (km/h)	
		Worst-user	Worst-IP	Expected	Worst	Expected
P_0 FR	walk	54.98	25.18	14.60	56.13	56.13
	wait	63.23	70.02	70.31		
	transit	97.50	74.37	74.05		
	total	215.71	169.57	158.96		
P_0 DR	wait	125.27	12.08	11.10	34.33	25.86
	transit		37.89	24.78		
	total		49.96	35.88		
P^* DR	wait	102.62	9.02	8.39	32.85	24.02
	transit		32.61	21.96		
	total		41.62	30.34		

Notes: FR shows walk, wait, transit and the total cost. DR removes walking. User cells are in minutes; provider cells are in km/h, not speed but the amortized total travel distance per hour. "Worst-user" reflects longest individual time.

Optimizing the partition through our distributionally robust framework yields P^* DR, which achieves the strongest user performance, with slight decrease of the worst provider cost from 34.33 km/h to 32.85 km/h with a similar decrease in expected provider cost. User wait and in-vehicle transit times continue to fall, with the maximum individual user cost reduced from 125.27 minutes in P_0 DR to 102.62 minutes.

The optimized partition shifts several district boundaries and, under the costs and constraints included in our model, identifies a different depot block group, as shown in Figure 9. This is a direct consequence of how the model balances routing distance, ODD-related effort, and demand uncertainty. It does not mean that the depot must be relocated in practice. Real systems often face additional considerations that fall outside the scope of our formulation, such as partition requirements, long-term facility leases, land availability, political priorities, or partnerships with existing sites. If these factors require the depot to remain at its current location, the decision maker can simply treat the depot block group as fixed. Our framework fully accommodates this by fixing the depot location in the optimization while leaving the rest of the procedure unchanged.

From these comparisons we can observe three takeaways. First, demand-responsive last-mile service provides immediate benefits even without modifying the existing partition. Routing vehicles to actual destination locations reduces unnecessary mileage and eliminates walking, which lowers both provider costs and user delays. Autonomous vehicles are particularly well suited to implement this service model because they can follow flexible routes without additional driver labor.

Second, optimizing the partition through distributionally robust design strengthens service reliability and fairness. Although the provider cost increases slightly, the optimized districts balance workloads more effectively and reduce the user times further. The robust partition anticipates spatial demand shifts and maintains consistent service quality across districts, which is essential for operators serving dispersed communities under demand ambiguities.

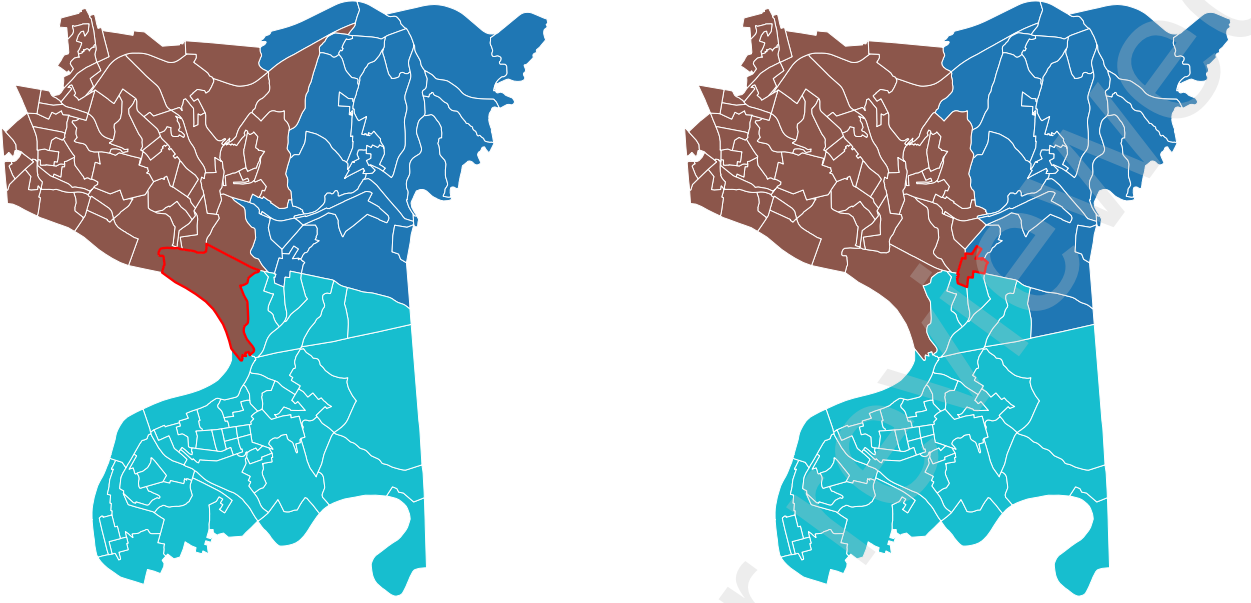


Figure 9: Partitions before and after Optimization

Left panel shows the original partition of fixed routes before optimization; right panel shows the optimal partition optimized against the worst-case distribution. The red outlined block group is the depot block group in each panel.

Third, AVs offer clear operational and sustainability advantages for demand-responsive last-mile service. They remove driver labor constraints, operate at low marginal cost, and follow routing plans with more accuracy. Their electric powertrains and precise maneuverability support lower energy use and reduced emissions compared with human-driven fleets. These attributes make AVs a manageable, cost-efficient, and environmentally sustainable platform for implementing flexible last-mile mobility services.

6. Conclusions

6.1. Generalizability of the Model and Solution Methodology

In summary, this paper demonstrates how ODD considerations and demand ambiguity can be jointly incorporated to optimize demand-responsive service design enabled by AVs. Our case study shows that demand-responsive AV service substantially reduces both provider travel cost and user waiting time relative to the original fixed-route design. Incorporating ambiguity-aware optimization further strengthens these gains by producing partitions, fleet sizes, and dispatch decisions that remain effective under shifting or uncertain demand patterns. Taken together, the results highlight the operational advantages of AVs, the performance benefits of demand-responsive service, and the value of integrating ambiguity-aware modeling into last-mile mobility planning.

This paper also provides a unified framework for designing AV-enabled last-mile services that batch demands spatial-temporally. In our specific setting, this framework jointly determines the depot location, service-region partition, fleet sizing, and dispatch frequency under spatial demand ambiguity and ODD constraints. At a high level, the planning problem has the structure

$$\min_{z \in \mathcal{Z}} \left\{ C_{\text{fixed}}(z) + C_{\text{ODD}}(z) + \sup_{f \in \mathcal{F}(z)} G(z, f) \right\},$$

where z collects all design variables (depot, partition, fleet size, dispatch interval), C_{fixed} captures deterministic system provider costs, C_{ODD} is the ODD-induced cost, and $G(z, f)$ is the service cost under demand density f drawn from an ambiguity set $\mathcal{F}(z)$. Formulation (1) makes this structure explicit for our case, while Figure 1 shows how the subsequent formulations transform it into a tractable robust partitioning problem. The case study instantiates this general template in a specific setting but the formulation itself is not tied to a particular city, fleet composition, or cost parameterization.

The components of the system cost can be modified without changing the optimization backbone. The ODD term $C_{\text{ODD}}(z)$ is represented through the modular cost $\mathcal{J}(\omega(P_i))$ in Formulation (1), where $\omega(P_i)$ summarizes environmental features of district P_i . Any other feature map or capability profile can be substituted in this term, for example to represent different AV platforms, human drivers, or mixed fleets, by redefining $\mathcal{J}$ while keeping the rest of the formulation unchanged. The overall service cost $G(z, f)$ is modeled in Formulation (2) using the BHH routing approximation in Appendix A.1 and the maximum cost across districts; this part can be replaced by alternative approximations that capture workload variance, energy use, emissions, or time-dependent congestion. Or we can also change G from the maximum cost across districts to the summation or even more complex functional of district-level costs.

Likewise, the ambiguity set $\mathcal{F}(z)$, represented as a Wasserstein ball in Formulations (3)–(4), can be redefined to use different metrics (by simply changing Euclidean distance d^{Euc} to others like Manhattan distance $d^{\text{Manhattan}}$, and hence changing parameters d_{ij} in (7)) or moment constraints by bounding the 1st and 2nd moments of distributions as per Delage and Ye (2010), i.e.,

$$\begin{aligned} \sum_{j=1}^J x_{ji}^2 &\leq 1, \quad x_{ji} \geq 0, \quad \forall i, j, \\ \sum_{j=1}^J x_{ji}^2 \xi_j &= \mu_i, \\ \sum_{j=1}^J x_{ji}^2 (\xi_j - \mu_{0,i})(\xi_j - \mu_{0,i})^\top &\leq \gamma_2 \Sigma_{0,i}, \\ \begin{bmatrix} \Sigma_{0,i} & \mu_i - \mu_{0,i} \\ (\mu_i - \mu_{0,i})^\top & \gamma_1 \end{bmatrix} &\geq 0, \end{aligned}$$

where the decisions x_{ji}, μ_i are the square root probability mass of block group B_j and the district P_i 's centroid weighted by probability mass; as parameters, ξ_j is the location of block group B_j , $\mu_{0,i}$ is the centroid weighted by the nominal distribution, $\Sigma_{0,i}$ is the estimated covariance matrix of district of P_i , and γ_0, γ_1 are bound scalars. These constraints still make the inner problem an SDP. It can be observed that such modifications preserve the overall structure

$$\min_{z \in \mathcal{Z}} \{C_{\text{fixed}}(z) + C_{\text{ODD}}(z) + \sup_{f \in \mathcal{F}(z)} G(z, f)\}.$$

The same framework also applies to multi-depot systems (by allowing $\delta_j = 1$ for multiple j 's), finer or coarser geographic units (by redefining the “blocks” as tract or streets), or designs with additional objectives such as equity or coverage guarantees (by adding more terms in the $G(z, f)$ or imposing cardinality constraints on $\{j : z_{ji} = 1\}$).

The solution method path in Figure 1 also highlights how the solution path generalizes. The distributionally robust motivation in Appendix A.2 and the BHH approximation in Appendix A.1 guide the construction of the conceptual model in Formulation (1). This model is then formalized in the precise continuous formulation (2), whose objective structure naturally leads to an inner DRO problem (Formulation (3)) for worst-case distribution. Introducing a Wasserstein ambiguity set refines this inner problem (3) into the infinite-dimensional formulation in (4). To make the problem tractable, Subsection 4.1 discretizes the continuous region and demand distributions with resulting $O(\delta)$ error bound proposed by Proposition 5, which implies that the resulting discrete approximation error diminishes as discrete blocks are refined. These discretizations convert the continuous inner problem (4) into the finite-dimensional primal and dual formulations (Formulations (6) and (5)), which in turn integrate directly into the unified discrete overall MINLP in Formulation (7), itself discretized using the same technique. Algorithms 1 and 2 operate on (7) by searching over depot locations, partitions, and optimizing subsequent fleet sizes, and dispatch intervals, thereby producing a high-quality design. Because each step transforms the problem in a structured way, moving from the conceptual model to continuous formulations, then to discrete versions, and finally into tractable forms, the overall solution path is modular and can be adapted to alternative operational modeling choices.

Therefore, this path is not specific to one setting. Other applications with a similar robust spatial-design structure can follow the same sequence of steps in Figure 1, replacing only the district-level cost measurement Φ with an alternative system-specific expression while keeping the remainder of the solution pathway unchanged. The approach does not depend on specific parameterizations and applies whenever service effort, waiting time, coverage, or other performance measures are defined over a continuous region. The resulting solution algorithm separates the search

procedure from the cost-evaluation subproblem, which makes it broadly applicable to spatial partition problems: different routing costs, batching rules, or robustness criteria can be incorporated by modifying only the cost-evaluation component, while the overall search and discretization framework remains intact. In this sense, Figure 1 provides a general map for translating a high-level robust design problem into a concrete, solvable discrete formulation. These forms of extensibility highlight the broader applicability of our modeling and solution approach, and we return to this theme when discussing potential extensions to other autonomy-aware or uncertainty-aware planning problems.

6.2. Future Directions

Our framework suggests several promising directions for extension. One way is to consider decision-dependent ambiguity sets, where the uncertainty in future demand depends on the chosen partitions or dispatch policies, creating a tighter coupling between design decisions and demand patterns. Another extension is to model ODD boundaries as dynamic rather than fixed, reflecting how sensor performance, infrastructure upgrades, or weather expectations may shift operating feasibility over time. Additionally, richer operational settings could also be studied by introducing multiple depots, heterogeneous AV capabilities, or mixed human–AV fleets, enabling broader exploration of deployment strategies. From a methodological view, combining our approach with online learning or adaptive estimation would allow ambiguity radii or cost parameters to update as new data arrives. Finally, integrating the framework with scenario-based safety assessments or high-fidelity simulators could bridge operations design with AV validation workflows, providing a more holistic tool for planning AV-enabled mobility systems.

A. Methodology Prerequisite Review

A.1. Continuous Approximations and Applications

A central methodological tool in vehicle routing and region design is the use of continuous approximations, which replace discrete demand samples by spatial densities to obtain tractable asymptotic expressions for routing length. A foundational result supporting this approach is the Beardwood–Halton–Hammersley (BHH) theorem, which characterizes the scaling of the optimal TSP tour through randomly located points (Beardwood et al., 1959; Steele, 1981). Unlike studies that assume uniform spatial demand, our setting requires nonuniform densities because demands concentrate around population centers, roadway structures, and other built-environment features. This nonuniformity is essential for incorporating spatial demand ambiguity through density-weighted routing approximations. The BHH theorem thus provides the mathematical link between a demand density and the resulting travel distance, forming the basis of the spatial cost components in our model.

Theorem 6 (Beardwood-Halton-Hammersley Theorem (Beardwood et al., 1959)). *Let $\mathbf{X}_1, \dots, \mathbf{X}_n, n \geq 1$ be i.i.d. random locations following the measure $\mathbb{P}$ with support P . Then the length L_n of a shortest TSP tour through the n locations satisfies*

$$\frac{L_n}{\sqrt{n}} \rightarrow \beta \int_P \sqrt{f(\mathbf{x})} d\mathbf{x}$$

with probability 1, as $n \rightarrow \infty$, where $f(\mathbf{x})$ is the density of the absolutely continuous part of $\mathbb{P}$ and β is a constant only depending on the metric space. In particular, $\beta \approx 0.7120$ when the metric space is $(\mathbb{R}^2, d^{Euc})$.

A.2. Distributionally Robust Optimization

Because routing length depends directly on the spatial demand density, which is ambiguous in practice, we employ distributionally robust optimization (DRO) to ensure reliable performance. DRO safeguards decisions against distributional ambiguity by minimizing the worst-case cost over an ambiguity set of plausible demand distributions (Delage and Ye, 2010; Ben-Tal, Den Hertog, De Waegenaere, Melenberg and Rennen, 2013). Unlike most existing DRO formulations, where the objective is an expected cost that is linear in the underlying density, the cost structure in our model depends nonlinearly on the density through the BHH TSP length approximation. For any decision vector θ , let $C(\theta, f)$ denote the total cost evaluated under density f . Given an ambiguity set $\mathcal{F}$ containing the plausible distributions, the DRO problem solves

$$\min_{\theta} \max_{f \in \mathcal{F}} C(\theta, f).$$

This formulation provides a performance guarantee when the true distribution $\underline{f}$ lies in the ambiguity set with high probability. In particular, if $\mathbb{P}(\underline{f} \in \mathcal{F}) \geq 1 - \delta$, then for any decision θ^{DRO} optimizing the worst-case cost we have

$$C(\theta^{\text{DRO}}, \underline{f}) \leq \max_{f \in \mathcal{F}} C(\theta^{\text{DRO}}, f) = \min_{\theta} \max_{f \in \mathcal{F}} C(\theta, f),$$

with probability at least $1 - \delta$. Thus the DRO solution optimizes a valid upper bound on the true operational cost. By optimizing against the worst-case distribution consistent with the data, DRO ensures that the resulting decision is robust to distributional ambiguities.

A.3. Wasserstein Distance and Ambiguity Set

Wasserstein-based DRO has become a prominent framework in transportation and service-system modeling because it preserves the spatial structure of demand distributions and explicitly accounts for geographic variability by relocating probability mass (Blanchet et al., 2019; Blanchet and Murthy, 2019; Gao and Kleywegt, 2023). This property is especially valuable in routing and region partitioning problems, where demand uncertainty manifests through spatial shifts. For two densities f and $\hat{f}$ defined on the service region S , the Wasserstein distance is

$$W(f, \hat{f}) = \inf_{\pi \in \Pi(f, \hat{f})} \int_{S \times S} d(\mathbf{x}, \mathbf{y}) d\pi(\mathbf{x}, \mathbf{y}),$$

where $\Pi(f, \hat{f})$ is the set of couplings with marginals f and $\hat{f}$, and $d(\mathbf{x}, \mathbf{y})$ denotes the ground metric on S . Intuitively, $W(f, \hat{f})$ measures the minimum effort required to transform one spatial distribution into the other, and this effort is

determined by geographic distance as captured by d . Because of this spatial interpretability, the Wasserstein distance offers a natural way to quantify distributional ambiguity arising from plausible user relocations or local demand shifts. Incorporating this spatial ambiguity into DRO yields the Wasserstein ambiguity set as

$$\mathcal{F}(\epsilon) = \{f \in L_1(S) : W(f, \hat{f}) \leq \epsilon\},$$

which represents all demand distributions that lie within a radius ϵ of the nominal estimate $\hat{f}$.

B. Worst-Case Distribution Solution Efficiency

A key computational bottleneck of distributionally robust service design is evaluating the worst-case demand distribution for each district. Existing methods, such as the analytic center cutting-plane method (ACCPM) in Carlsson et al. (2018), are theoretically precise and elegant but hard to generalize tractably to our problem class. In this subsection, we compare the ACCPM with a direct solution of our discrete CQCP formulation (6) to illustrate how structural differences translate into substantial practical efficiency gains.

The two approaches scale with fundamentally different problem-size measures. The ACCPM's runtime grows with the number of demand points, because each analytic center update solves a nonlinear problem whose size depends directly on the point set. In contrast, our direct-solve method scales primarily with the chosen grid resolution. Once the resolution is fixed, the problem size remains stable regardless of how many demand points appear in the region. This distinction has practical implications. The number of observed demand points is determined by the data and cannot be controlled by the decision maker. When the dataset contains many demand points, the ACCPM faces an unavoidable disadvantage because its runtime increases steeply with the sample size. On the other hand, the grid resolution in (6) is a modeling choice that can be set to a practical and computationally manageable level. This independence from the sample size is inherent to the formulation and leads to much more stable runtimes when the number of demands is large or uncertain.

These structural differences manifest clearly in runtime behavior. As shown in Figure 10 (left panel), the ACCPM exhibits steep growth in runtime as the number of demand points increases and can experience time out even for moderate point sets. In contrast, solving the CQCP directly using Gurobi scales smoothly with resolution and remains tractable for fine grids. This enables us to push the discretization to higher resolutions to capture finer spatial variation without experiencing prohibitive computation times. In addition to the runtime considerations, the right panel of Figure

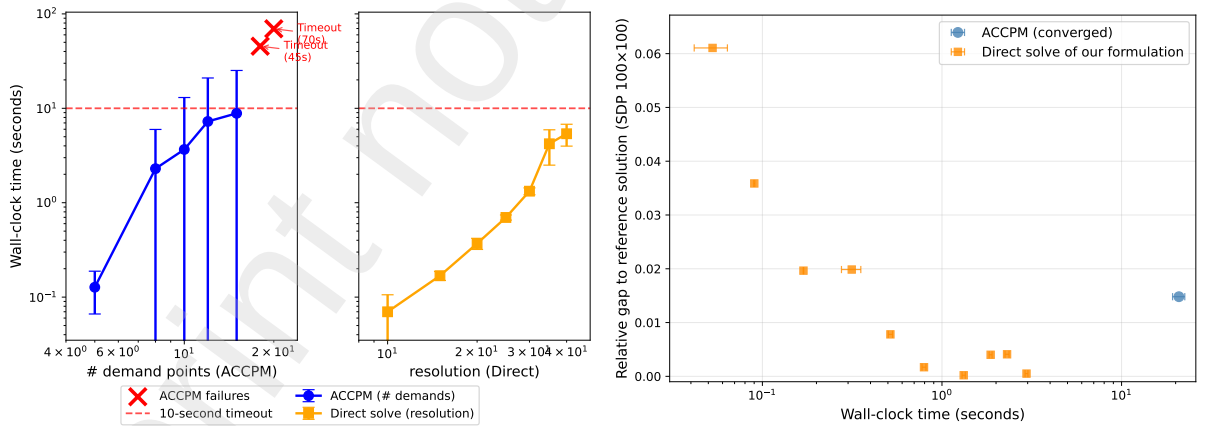


Figure 10: Scalability: ACCPM vs Direct Solving the CQCP (different size measures)

The left panel reports the computational times of ACCPM and Gurobi when solving the CQCP. ACCPM's runtime increases steeply with # of demands and eventually encounters a numerical failure, while directly solving the CQCP via Gurobi exhibits a more moderate growth in runtime. The right panel illustrates the trade-off between runtime and solution accuracy. For the instance shown, ACCPM achieves an error of about 1.5% but requires a relatively long runtime, whereas the direct-solve approach attains shorter runtimes and can improve the accuracy to below 1% by using higher-resolution discretizations.

10 shows differences in numerical accuracy between the two approaches. Although the ACCPM is theoretically an

exact continuous solver, its practical performance is undermined by several numerical challenges: repeated integral evaluations introduce approximation errors in floating-point arithmetic, maintaining high precision for nonlinear equality-constrained subproblems becomes increasingly difficult over many iterations, and the method is sensitive to parameter tuning and analytic-center computations, often exhibiting fluctuations before convergence. These issues can result in noticeable deviations from the reference solution and, in some cases, prevent the ACCPM from recovering the worst-case distribution within the time limit.

By contrast, directly solving our CQCP (6) produces rapid and near-exact solutions. For the discretizations typically used in design optimization, the relative error remains within a few percent while requiring only a fraction of the ACCPM runtime. Moreover, by increasing the grid resolution, our method can achieve even higher accuracy than the ACCPM while still completing in significantly less time. This trade-off illustrates the practical advantage of the direct-solve approach: in real decision-making contexts, obtaining a high-quality approximation quickly is often far more valuable than computing a theoretically exact solution at substantially greater computational cost.

C. Evaluating $\Phi(D, S, \mathbf{z})$

Given $D, S, \mathbf{z}$ as the input of Φ , the depot location, partition, and roots are determined. Hence, the depot cost $\sum_{j=1}^N \phi(\kappa_j) \delta_j$, ODD cost O , and each district's linehaul cost K_i can be calculated accordingly. It remains for each district P_i to 1) find the worst-case distribution captured by $\{x_{ji}^*\}$, and 2) optimize the fleet size V_i and dispatch interval T_i . Given the bilinearity between $\alpha(P_i)$ (defined in (4)) and its coefficient (shown in Subsection 4.2), we can solve (6) to obtain the approximated $\hat{\alpha}(\hat{P}_i)$ as well as the worst-case distribution $\{x_{ji}^*\}$. Then the problem becomes the optimization of (8) (meanwhile we need to keep V_i as low as possible because of the positive per-AV cost l), which is derived by decomposing relevant objective terms, decisions and constraints from (7).

$$\min_{T_i, V_i} \frac{K_i}{T_i/V_i} + \hat{\alpha}(\hat{P}_i) \left(\frac{1}{\sqrt{T_i/V_i}} + \frac{w_r}{s} \sqrt{T_i/V_i} \right) + w_r T_i/V_i \quad (8a)$$

$$\text{s.t.} \quad \frac{T_i}{V_i} \leq \frac{\bar{\lambda}(\mathcal{Q}, \epsilon)}{\Lambda \sum_{j=1}^N (x_{ji}^*)^2} \quad (8b)$$

$$\bar{Q}_i = \frac{(sT_i - K_i)^2}{\beta^2} \frac{\sum_{j=1}^N (x_{ji}^*)^2}{\left(\sum_{j=1}^N x_{ji}^*\right)^2} \quad (8c)$$

$$\frac{T_i}{V_i} \leq \frac{\bar{\lambda}(\bar{Q}_i, \epsilon)}{\Lambda \sum_{j=1}^N (x_{ji}^*)^2} \quad (8d)$$

Note in (8), we omit fixed constant terms (determined by $D, S, \mathbf{z}$) for simplicity. And T_i, V_i appear together in $C_i = T_i/V_i$. By the Hessian of (8a) on $\sqrt{T_i/V_i}$, (8a) is strictly convex in $\sqrt{T_i/V_i}$ (actually U-shaped on $\sqrt{T_i/V_i}$), hence we can obtain the unconstrained minimizer $C_i^* := T_i/V_i$ to (8a) by taking the first order condition. Then we consider the constraint (8b), which essentially imposes an upper bound on T_i/V_i because $\mathcal{Q}, \epsilon, \Lambda$ are all parameters while x_{ji}^* 's have been fixed. Given the U-shape, if $C_i^* \leq \bar{\lambda}(\mathcal{Q}, \epsilon)/\Lambda \sum_{j=1}^N (x_{ji}^*)^2$, (8b) is already satisfied; otherwise if $C_i^* > \bar{\lambda}(\mathcal{Q}, \epsilon)/\Lambda \sum_{j=1}^N (x_{ji}^*)^2$, the optimal C_i^* should be updated as $\bar{\lambda}(\mathcal{Q}, \epsilon)/\Lambda \sum_{j=1}^N (x_{ji}^*)^2$.

We have determined the ratio $C_i^* T_i/V_i$, it remains to determine a lowest feasible V_i^* satisfying (8c) and (8d) and then T_i^* is inferred as $C_i^* V_i^*$. To do this, notice $\bar{Q}_i$ is strictly increasing in T_i and $\bar{\lambda}(\bar{Q}_i, \epsilon)$ (defined in Proposition 2) is also strictly increasing in $\bar{Q}_i$. Hence, sufficiently large V_i (hence sufficiently large $T_i = C_i^* V_i$) will satisfy both constraints, but we need to keep V_i as low as possible to minimize the AV fleet fixed cost, i.e. we are looking for an integer V_i^* such that

$$\bar{Q}_i = \frac{(sC_i^* V_i^* - K_i)^2}{\beta^2} \frac{\sum_{j=1}^N (x_{ji}^*)^2}{\left(\sum_{j=1}^N x_{ji}^*\right)^2}, \bar{Q}_i' = \frac{(sC_i^* (V_i^* - 1) - K_i)^2}{\beta^2} \frac{\sum_{j=1}^N (x_{ji}^*)^2}{\left(\sum_{j=1}^N x_{ji}^*\right)^2}, \frac{\bar{\lambda}(\bar{Q}_i', \epsilon)}{\Lambda \sum_{j=1}^N (x_{ji}^*)^2} < C_i^* \leq \frac{\bar{\lambda}(\bar{Q}_i, \epsilon)}{\Lambda \sum_{j=1}^N (x_{ji}^*)^2},$$

and this can be done by binary search over the potential range of V_i . Plugging T_i^*, V_i^* back yields Φ .

D. Additional Proofs

Proof of Proposition 5. Since f is Lipschitz on the compact region S , f is continuous and bounded. Denote the bound and the Lipschitz constant as

$$M_{\sqrt{f}} = \sup_{\mathbf{x} \in S} \sqrt{f(\mathbf{x})} < \infty, \quad L_{\sqrt{f}} = \sup_{\mathbf{x} \neq \mathbf{y}} \frac{|\sqrt{f(\mathbf{x})} - \sqrt{f(\mathbf{y})}|}{\|\mathbf{x} - \mathbf{y}\|} < \infty.$$

Then by triangle inequality, the difference between the precise α_i and the approximated $\hat{\alpha}_i$ can be bounded by the summation of the partition error and the distribution error

$$\left| \alpha(P_i) - \hat{\alpha}_i(\hat{P}_i) \right| \leq \underbrace{\left| \alpha(P_i) - \alpha(\hat{P}_i) \right|}_{\text{(I) Partition error}} + \underbrace{\left| \alpha(\hat{P}_i) - \hat{\alpha}_i(\hat{P}_i) \right|}_{\text{(II) Distribution error}}$$

(I) Partition error. Since $\alpha(\hat{P}_i) = \beta \sqrt{\Lambda} \int_{\hat{P}_i} \sqrt{f}$, we have $|\alpha(P_i) - \alpha(\hat{P}_i)| = \beta \sqrt{\Lambda} \int_{P_i \Delta \hat{P}_i} \sqrt{f(\mathbf{x})} d\mathbf{x} \leq \beta \sqrt{\Lambda} M_{\sqrt{f}} |P_i \Delta \hat{P}_i|$. Since $P_i \Delta \hat{P}_i \subset \{\mathbf{x} : \text{dist}(\mathbf{x}, \partial P_i) \leq \delta\}$, so $|P_i \Delta \hat{P}_i| \leq |\partial P_i| \delta + O(\delta^2)$, where $|\partial P_i|$ is the one-dimensional measure of the boundary. Hence, we have $|\alpha(P_i) - \alpha(\hat{P}_i)| = O(M_{\sqrt{f}} |\partial P_i| \delta)$.

(II) Distribution error. On each block $B_j \subset \hat{P}_i$, let $m_j = \int_{B_j} f$ and write $\tilde{f}(\mathbf{x}) = m_j / |B_j|$. By the intermediate-value theorem there exists $\boldsymbol{\eta}_j \in B_j$ with $\sqrt{f(\boldsymbol{\eta}_j)} = \sqrt{m_j / |B_j|}$, so $\int_{B_j} \sqrt{f} = \int_{B_j} \sqrt{f(\boldsymbol{\eta}_j)} d\mathbf{x}$. Since $\sqrt{f}$ is $L_{\sqrt{f}}$ -Lipschitz and $\text{diam}(B_j) \leq \delta$,

$$\left| \int_{B_j} \sqrt{f} - \int_{B_j} \sqrt{\tilde{f}} \right| \leq \int_{B_j} L_{\sqrt{f}} \delta d\mathbf{x} = L_{\sqrt{f}} \delta |B_j|.$$

Summing over all $B_j \subset P_i$ gives $|\alpha(\hat{P}_i) - \hat{\alpha}_i(\hat{P}_i)| = O(L_{\sqrt{f}} |S| \delta)$. Combining (I) and (II) yields $|\alpha_i - \hat{\alpha}_i| = O(M_{\sqrt{f}} |\partial P_i| \delta + L_{\sqrt{f}} |S| \delta) = O(\delta)$. $\square$

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